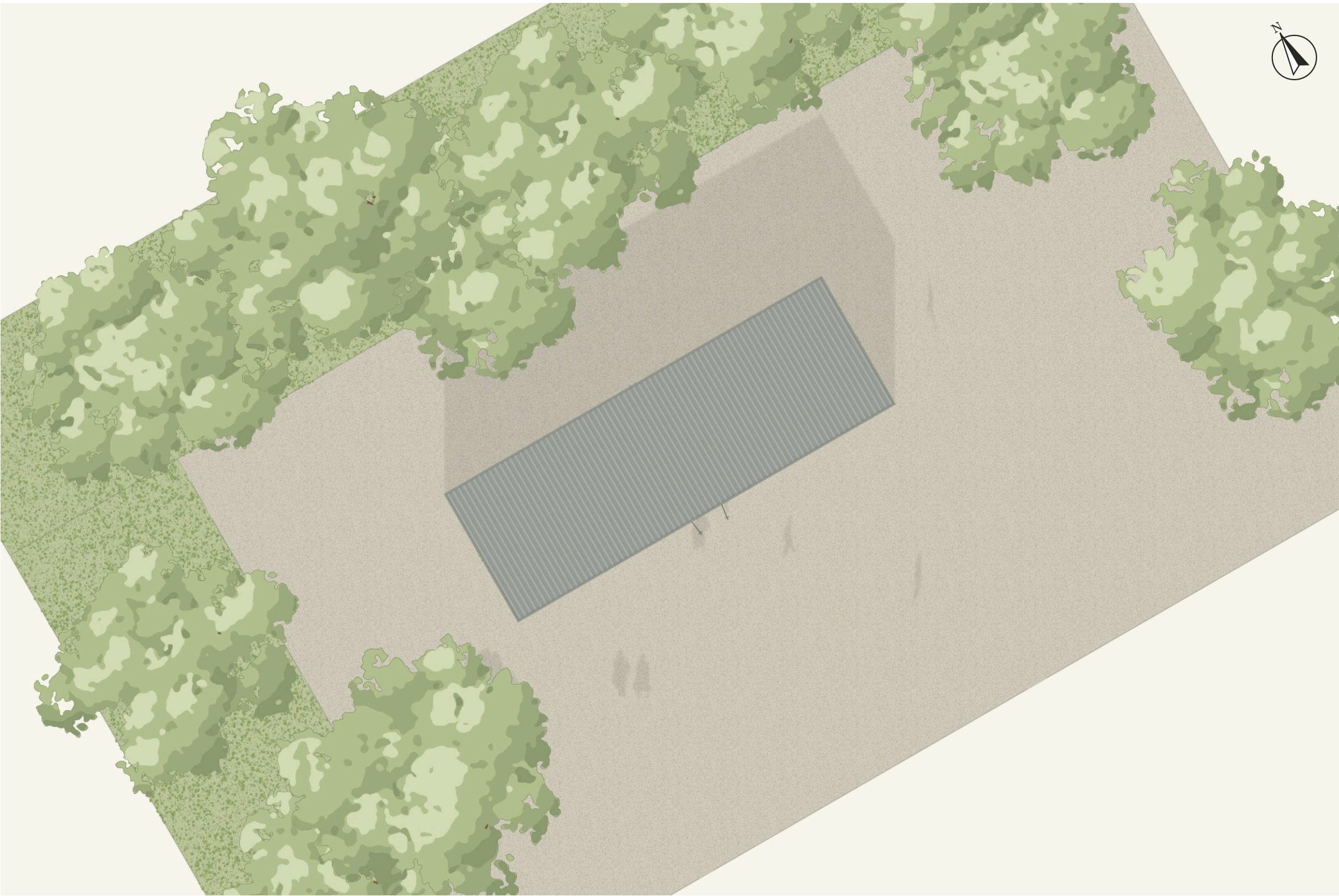




1 Axonometric from the south-east day - site, trees and library people - straight from the model



2 Site plan Museumpark setting (interpreted) - roof, trees seen from above

0 2 4 6 8 10 m - 1:150

About the drawings

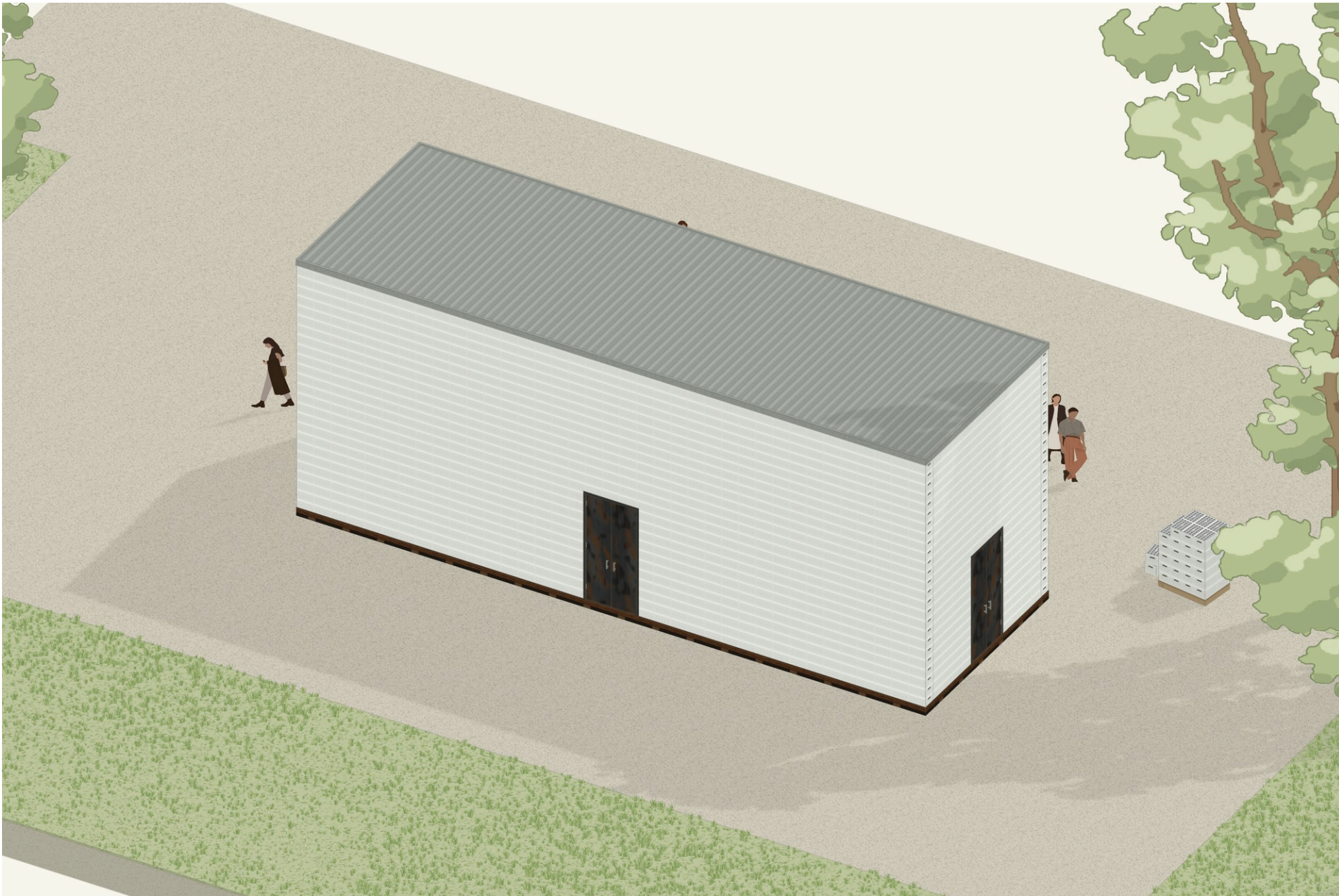
Atelier Kempe Thill's Light Building: a 15 × 5 × 6 m museum pavilion stacked from 2,268 colourless Dutch beer crates, tied by steel rods and erected by six people in one day.

All views are rendered from one Rhino model (v6, metres) in Three-Vis; plans and sections are cut from the same model.

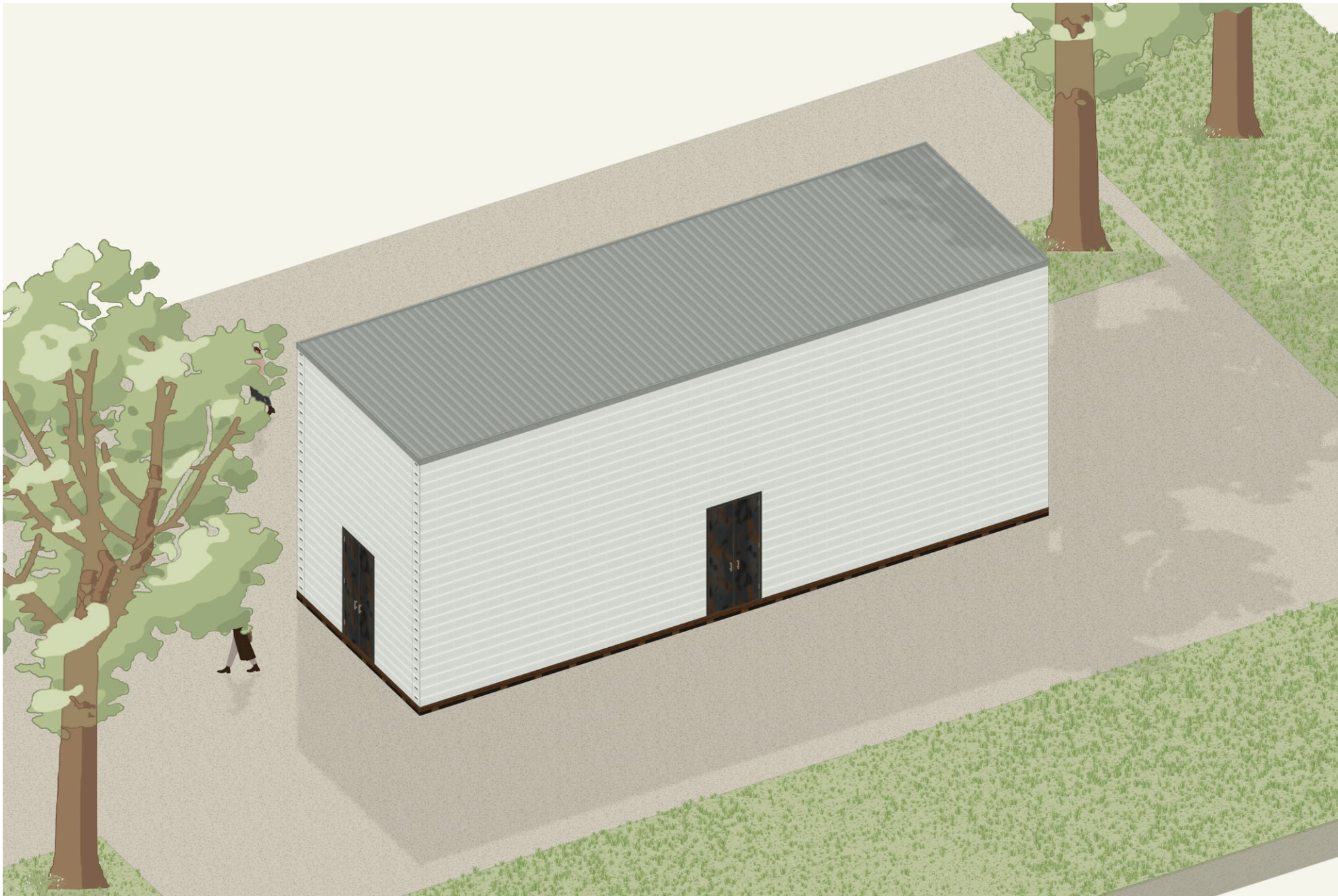
Measured (photo, plan, section): crate 395 × 295 × 235, 37 × 11 crates, 25 courses, doors 3 × 11 courses, levels. Interpreted: tie rods, clamp plates, channel, sleepers, packing blocks, anchors, joists. Invented: site, trees, people, the crate stack, the open south door.



3 Axonometric from the south-west day - with site



4 Axonometric from the north-west day - with site - the poplar row in front left out



5 Axonometric from the north-east day - with site - the poplar row in front left out

Light Building

Museum pavilion of translucent beer crates - Museumpark, Rotterdam 2001

ARCHITECT
Atelier Kempe Thill

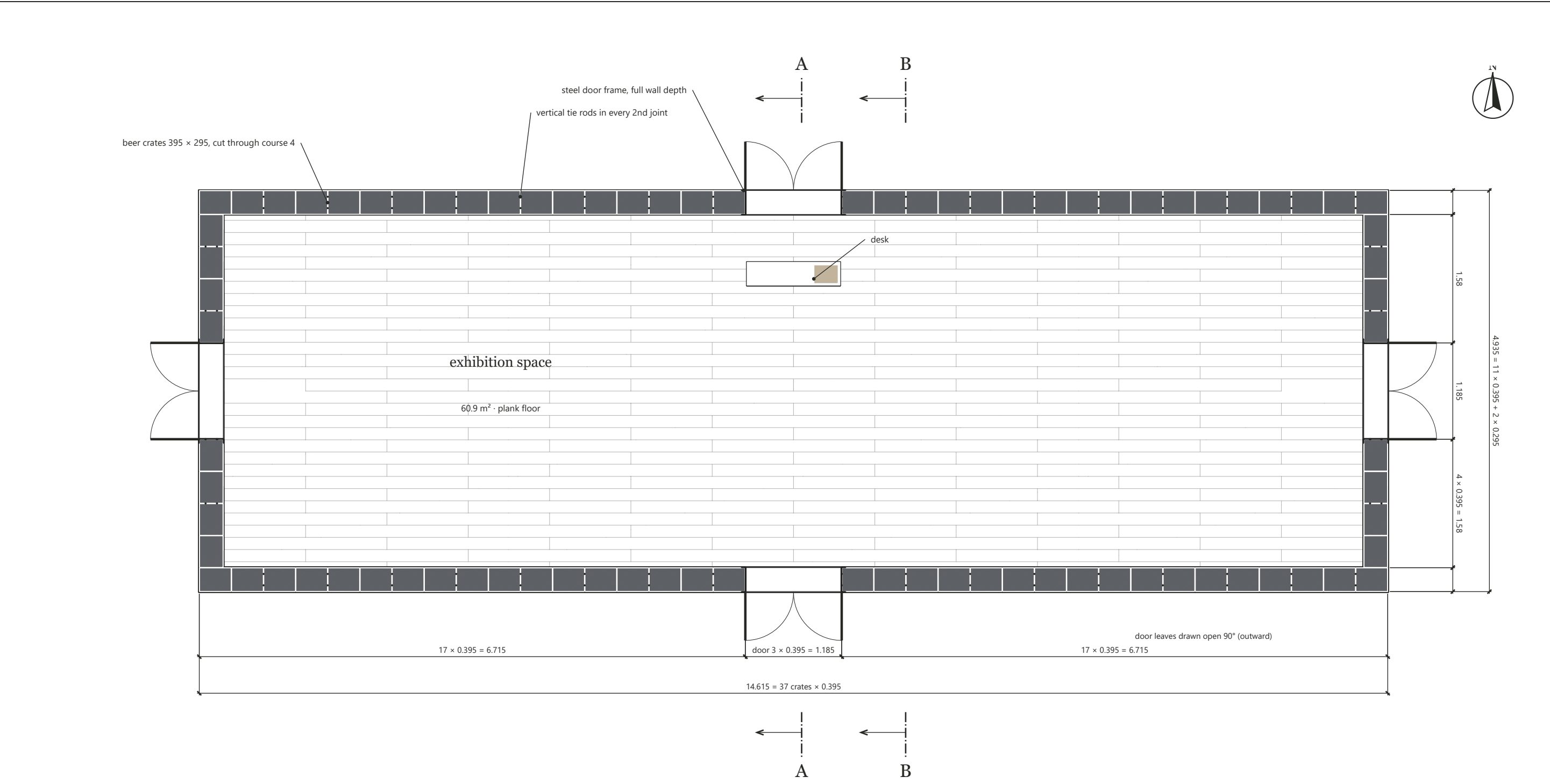
DRAWING
Axonometrics and site

SCALE
site 1:150

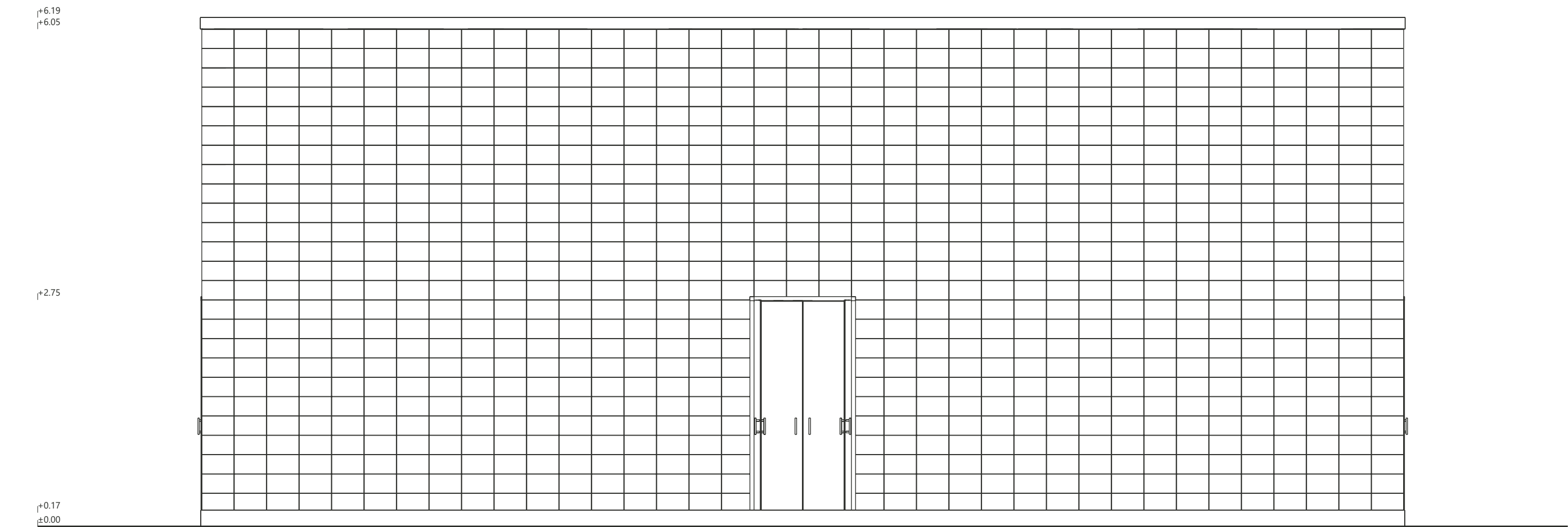
SHEET
01 / 08

DATE
4 October 2026

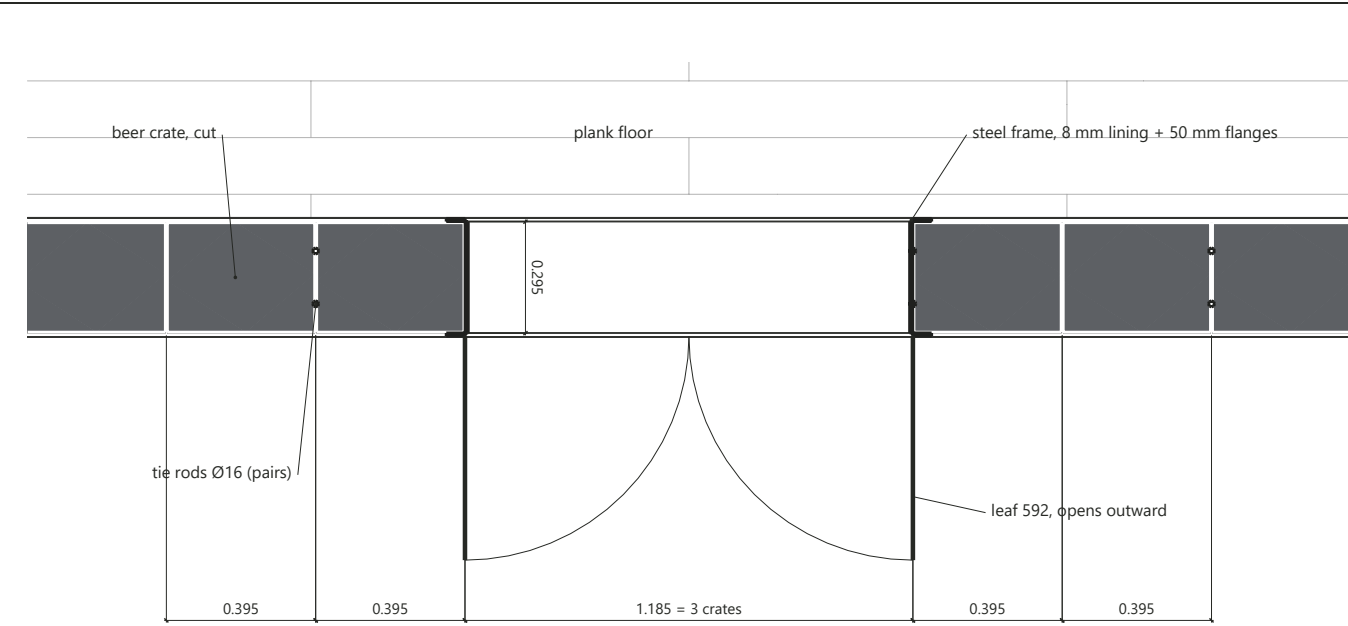
DRAWN
made with Three-Vis
from model v6, metres



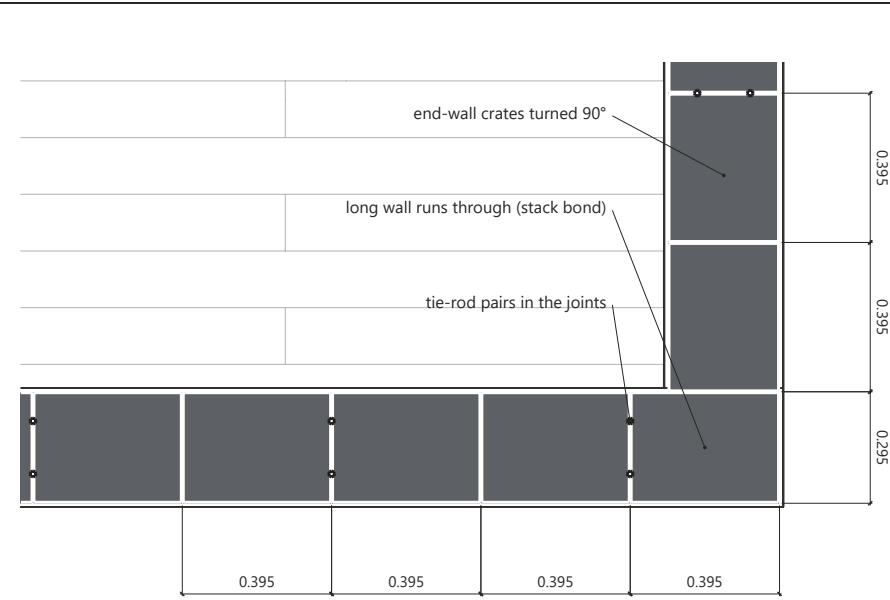
1 Floor plan, cut at +1.00 cut crates in stack bond · doors drawn open 90° (outward)



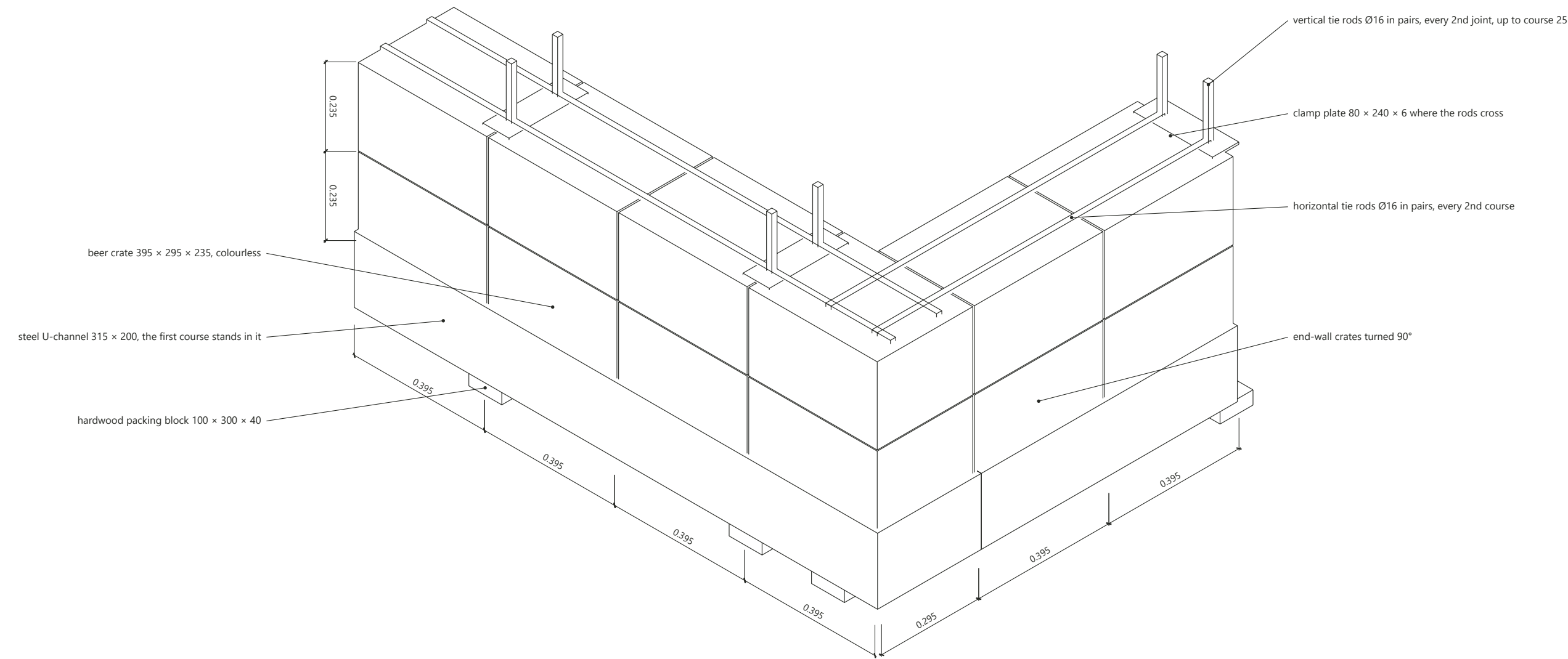
2 South elevation ink, every crate edge from the model · south door open



4 South door plan detail · frame, leaves, tie rods



5 South-east corner plan detail · stack bond

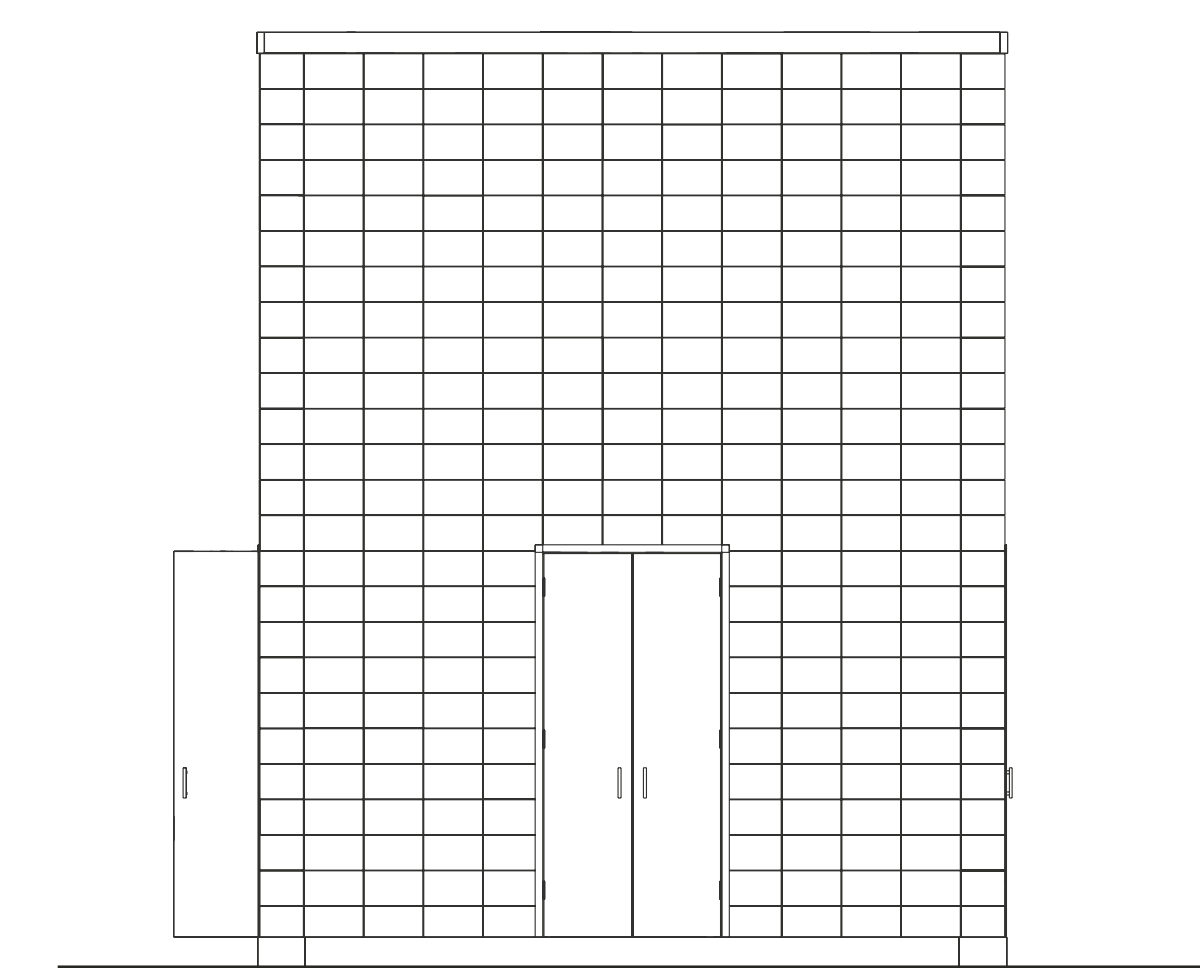


6 Crate module, isometric the south-east corner, courses 1–2, as it is tied: rods in pairs, clamp plates, channel on packing blocks

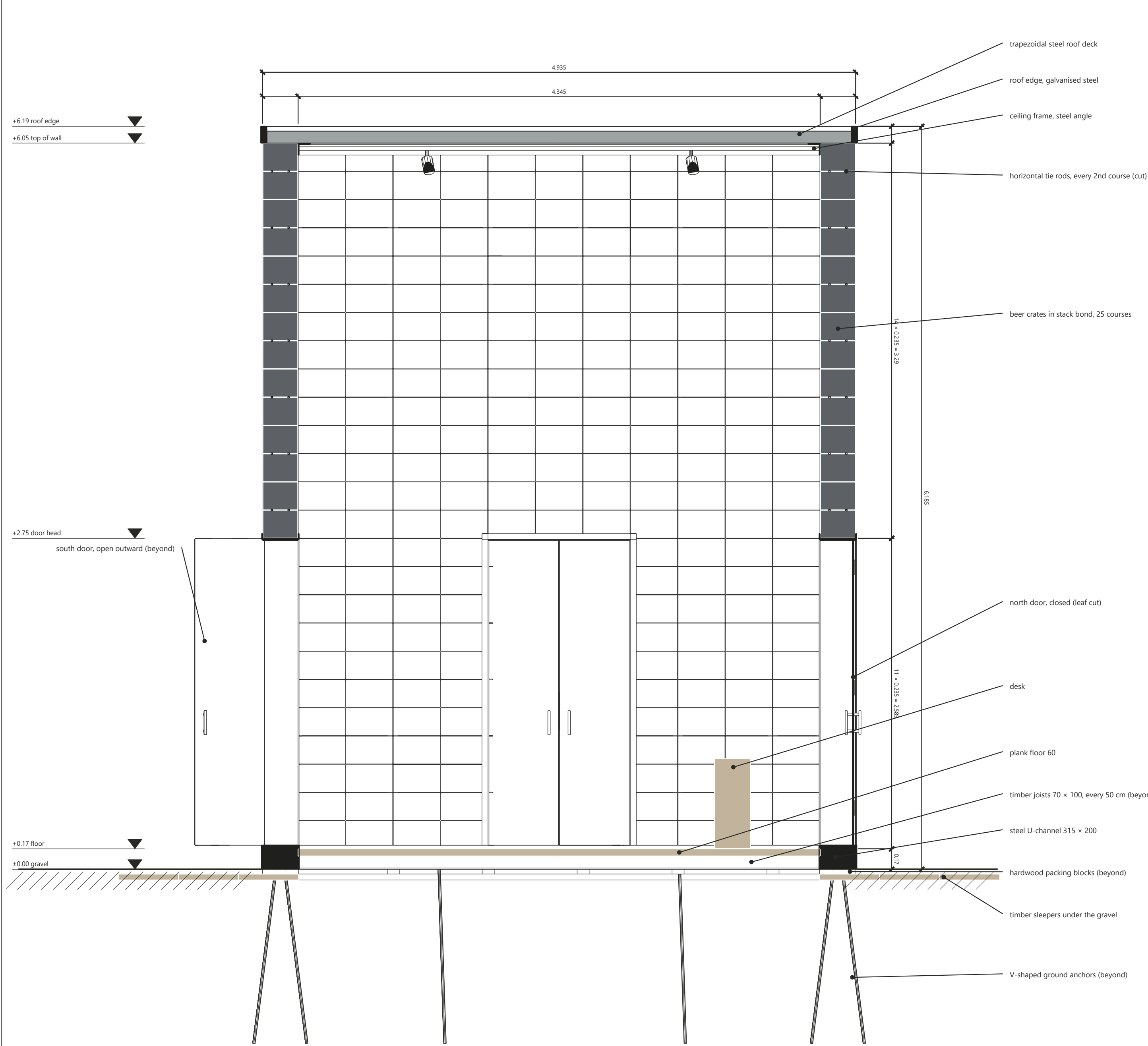
Plan notes

Crate 395 × 295 × 235 mm (Dutch standard crate, colourless edition), stacked in stack bond: every course the same, the long walls run through the corners.
Long walls 37 crates = 14.615 m (34 in courses 1–11, beside the door); end walls 11 crates between the long walls (8 beside the door); 4.935 m overall. 25 courses, top of wall +6.05.
Four door pairs, 3 crates × 11 courses = 1.185 × 2.585 m, in full-depth steel frames, opening outward. The desk stands at the north door.
Dimensions in metres. Section marks A and B refer to sheet 03.

beer crates, cut steel, cut timber, cut packing block, cut

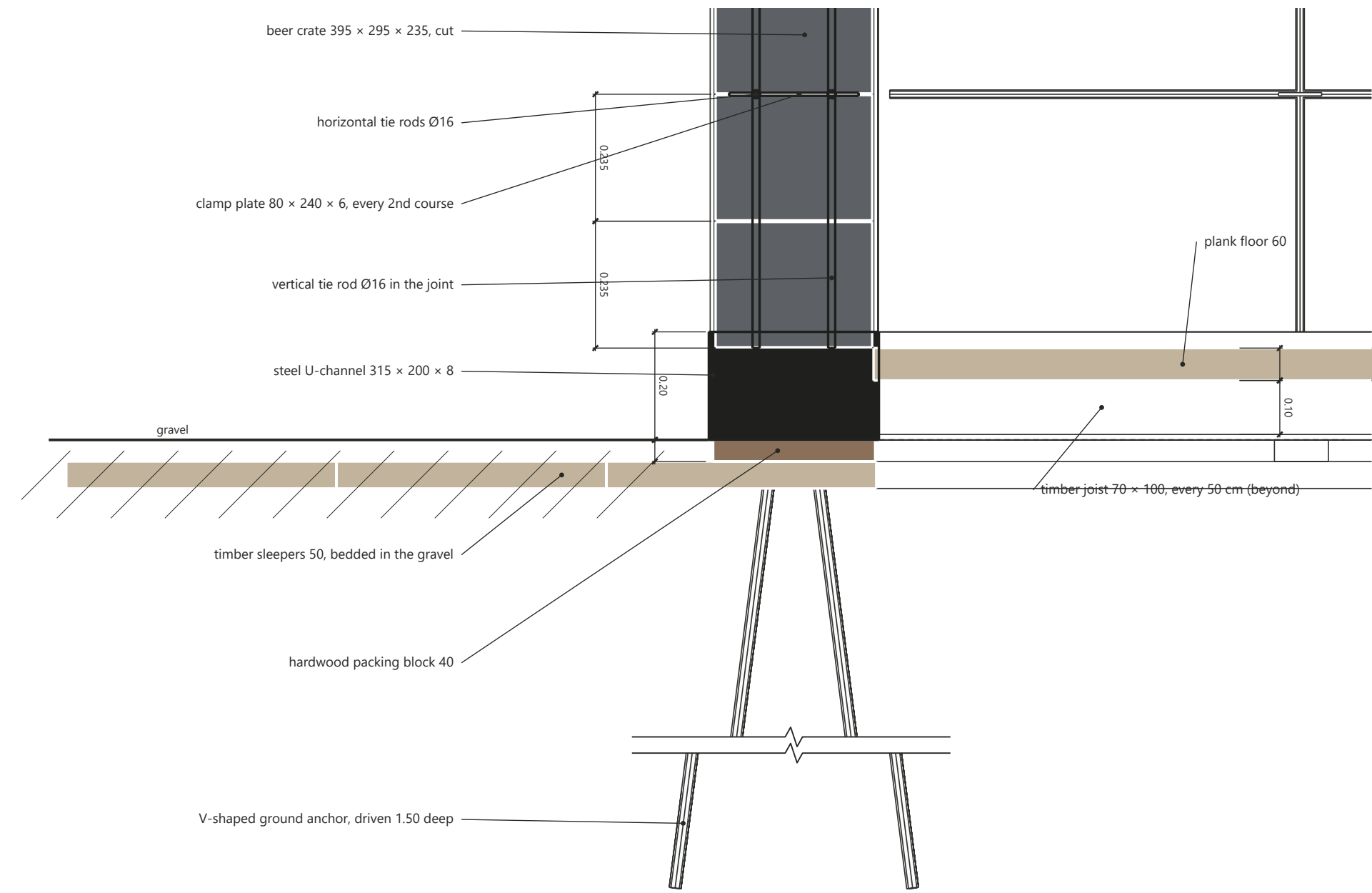


3 East elevation south door leaf open on the left

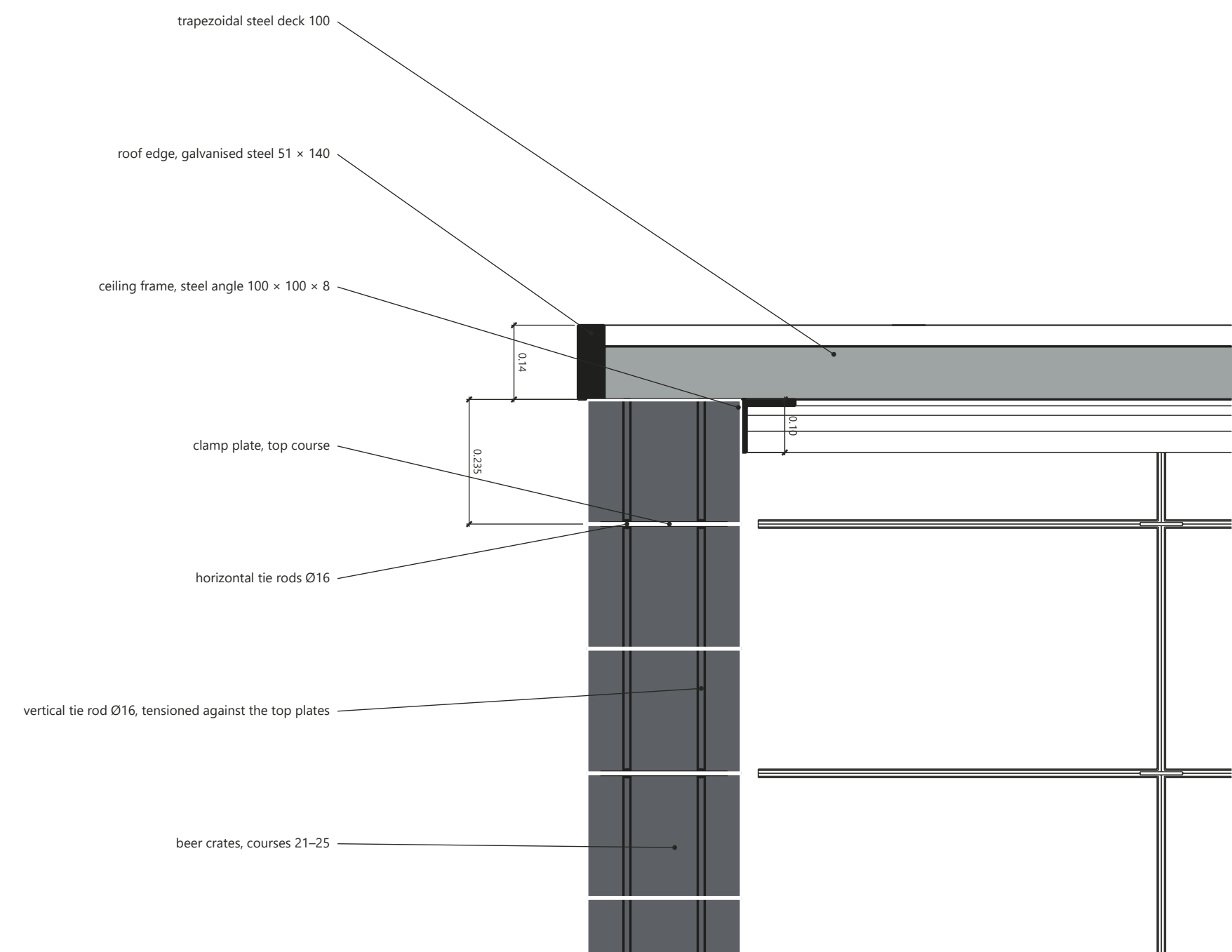


1 Section A-A, north-south through the doors looking west - cut at x = 0.10 m

0 0.5 1 1.5 2 m - 1:25



2 Wall base, section B-B channel on packing blocks, sleepers bedded in the gravel - anchors broken



3 Wall head, section B-B roof deck, edge and ceiling frame on the top course

0 0.1 0.2 0.3 0.4 0.5 m - 1:10

Build-up

Roof: trapezoidal steel deck 100 in a galvanised edge 51 × 140, on a steel angle frame along the inner wall face (+5.95 to +6.05).
Wall: 25 courses of beer crates in stack bond; vertical tie rods Ø16 in pairs in every 2nd joint, horizontal rods every 2nd course, clamp plates where they meet.
Floor: planks 60 on timber joists 70 × 100 at 50 cm, laid inside the channel ring (+0.17).
Base: steel U-channel 315 × 200 on hardwood packing blocks 40 (one under every rod position), on timber sleepers bedded at the underside of the gravel; driven V-anchors hold the channel down.

Open question resolved

The section drawing shows timber sleepers under the gravel; the site photo shows the rust channel resting on small blocks above a dark gap. Both hold: the sleepers lie under the gravel and the packing blocks level the channel on them (model v6). All levels above ±0.00 stay as drawn.



1 Eye level from the south-east 1.6 m · with site and library people



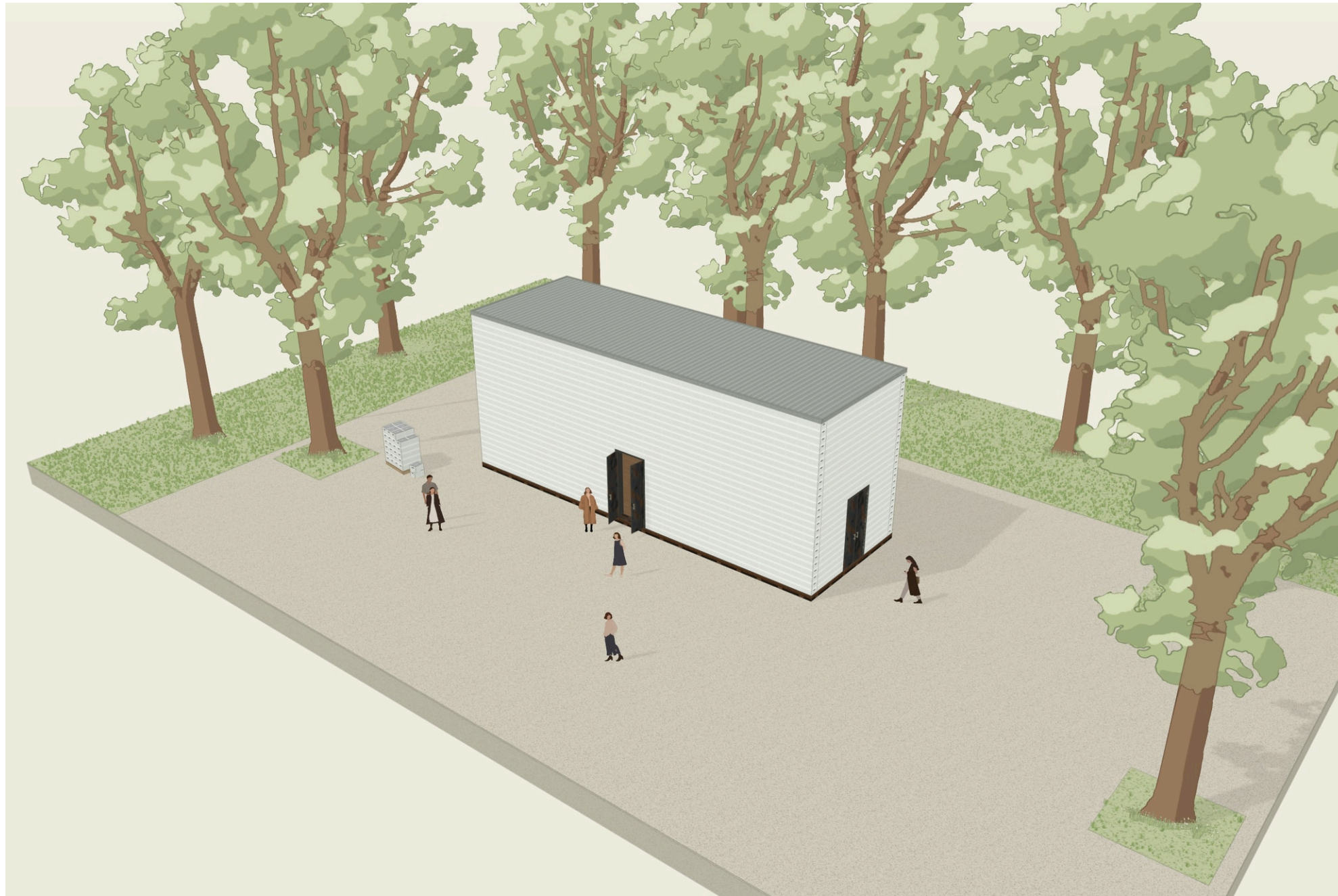
2 Eye level from the north-west 1.6 m



3 At the south door 1.6 m · approaching the open door



4 South facade, framed as the reference photo level camera 28 m south at 1.6 m



5 Aerial from the south-east



6 Aerial from the south-west

Only the building — the same cameras without site, trees and people



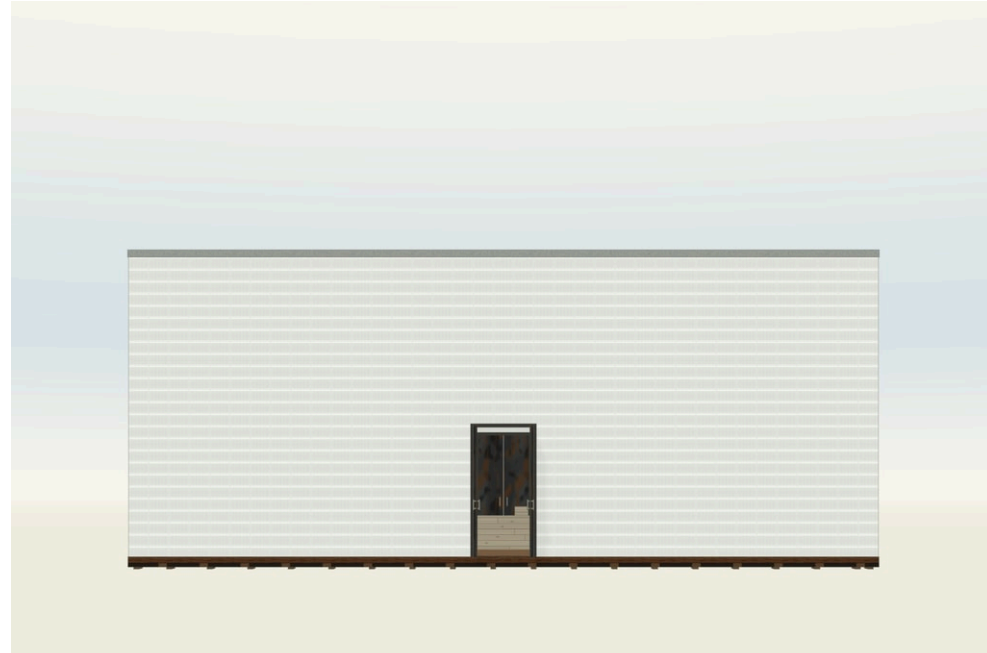
7 eye level, south-east



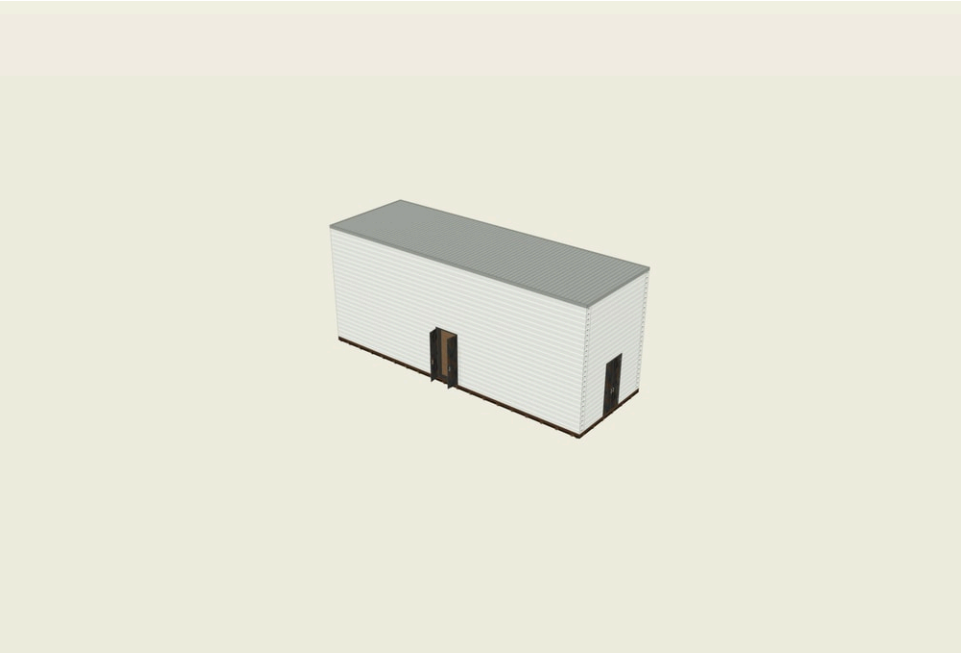
8 eye level, north-west



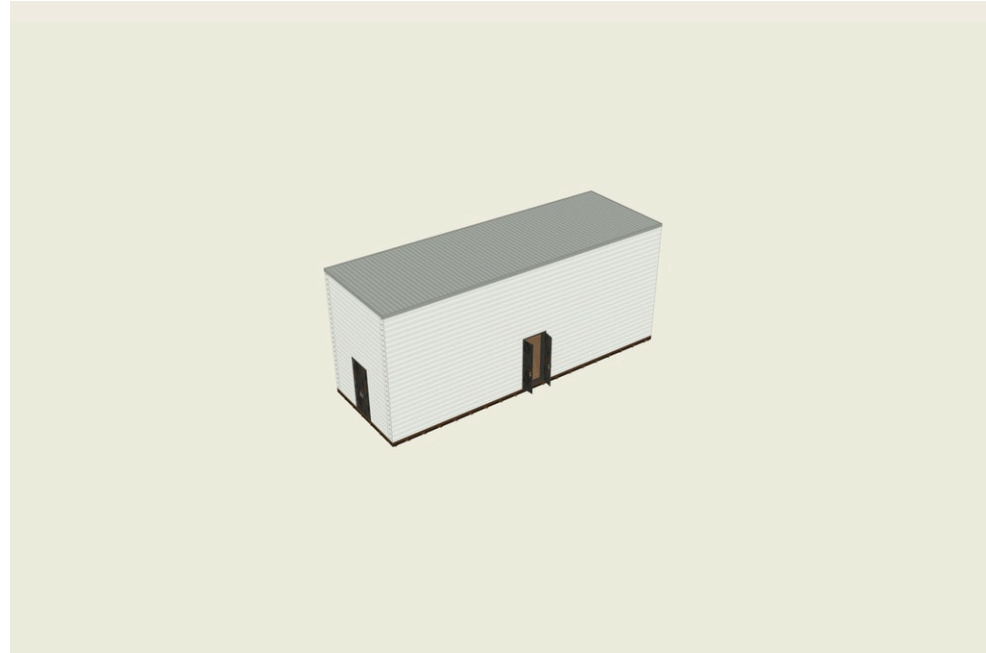
9 south door



10 south facade



11 aerial, south-east



12 aerial, south-west

Light Building

Museum pavilion of translucent beer crates · Museumpark, Rotterdam 2001

ARCHITECT
Atelier Kempe Thill

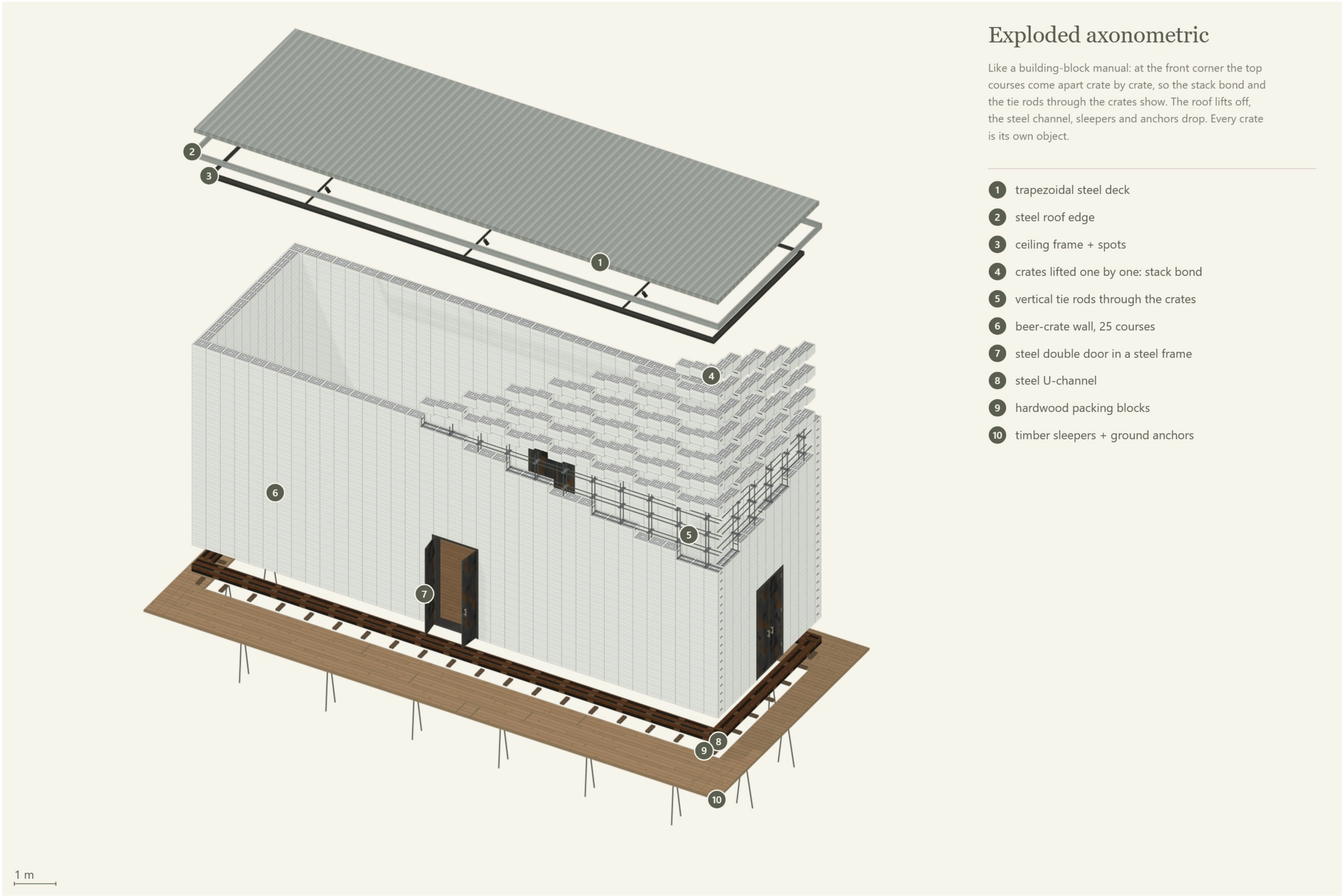
DRAWING
Perspectives

SCALE
perspective

SHEET
04 / 08

DATE
4 October 2026

DRAWN
made with Three-Vis
from model v6, metres

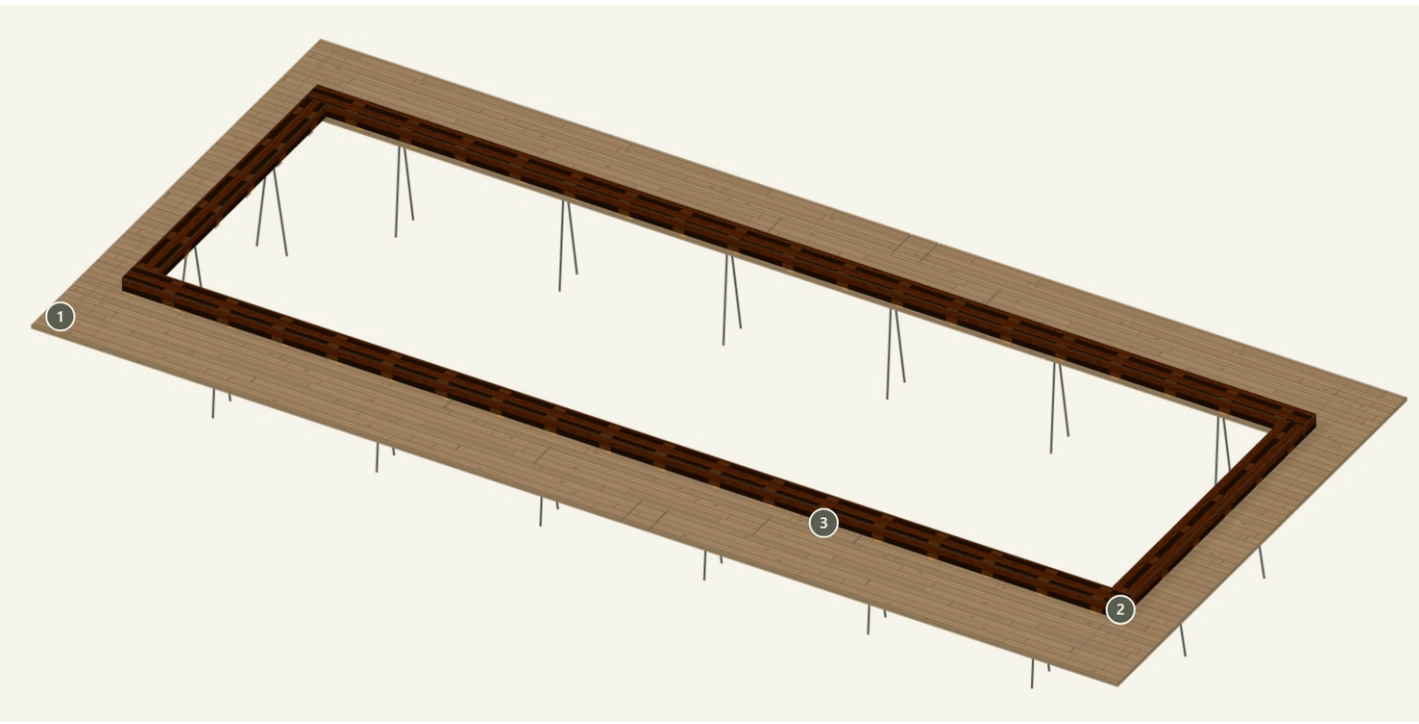


1 Exploded axonometric every crate its own object - roof lifted, base dropped

Exploded axonometric

Like a building-block manual: at the front corner the top courses come apart crate by crate, so the stack bond and the tie rods through the crates show. The roof lifts off, the steel channel, sleepers and anchors drop. Every crate is its own object.

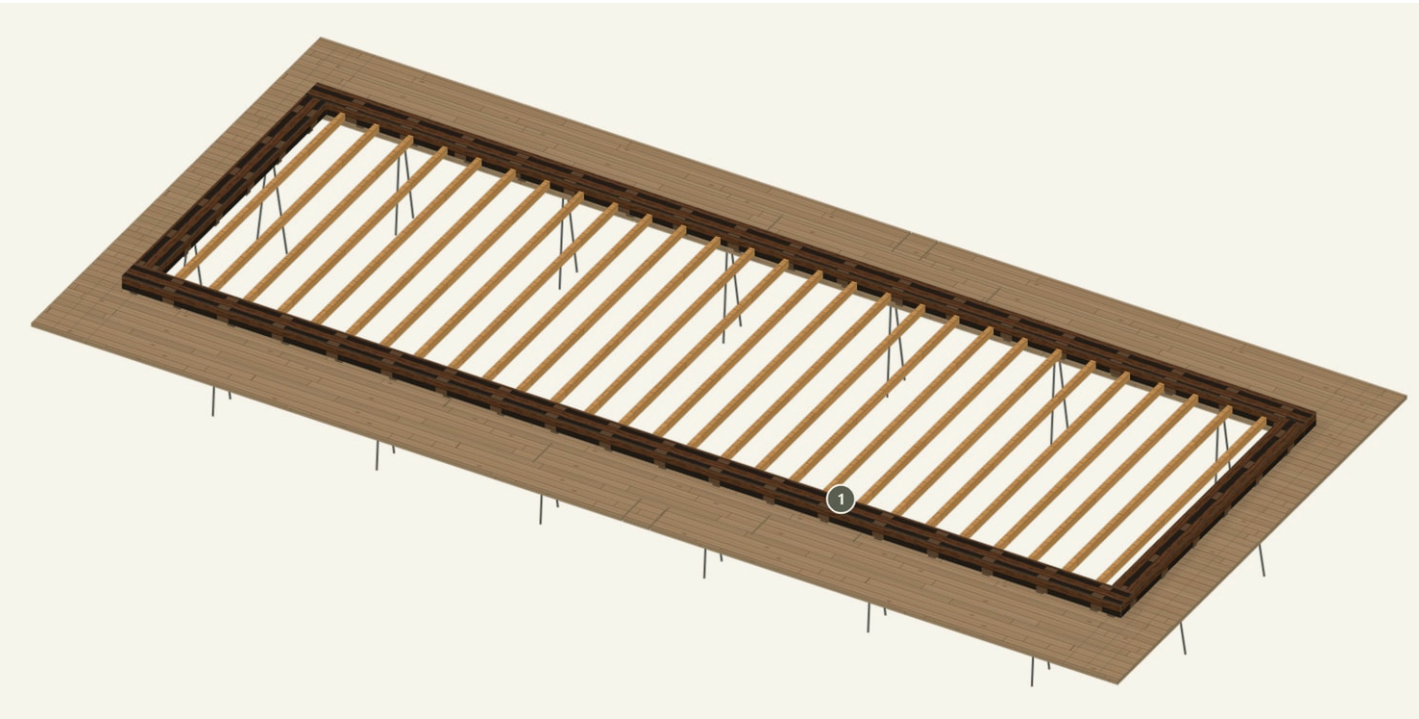
- 1 trapezoidal steel deck
- 2 steel roof edge
- 3 ceiling frame + spots
- 4 crates lifted one by one: stack bond
- 5 vertical tie rods through the crates
- 6 beer-crate wall, 25 courses
- 7 steel double door in a steel frame
- 8 steel U-channel
- 9 hardwood packing blocks
- 10 timber sleepers + ground anchors



2 Step 1 Foundation

Timber sleepers bedded in the gravel spread the load; hardwood packing blocks level the rusty steel channel that takes the first course, and driven V-anchors hold it down.

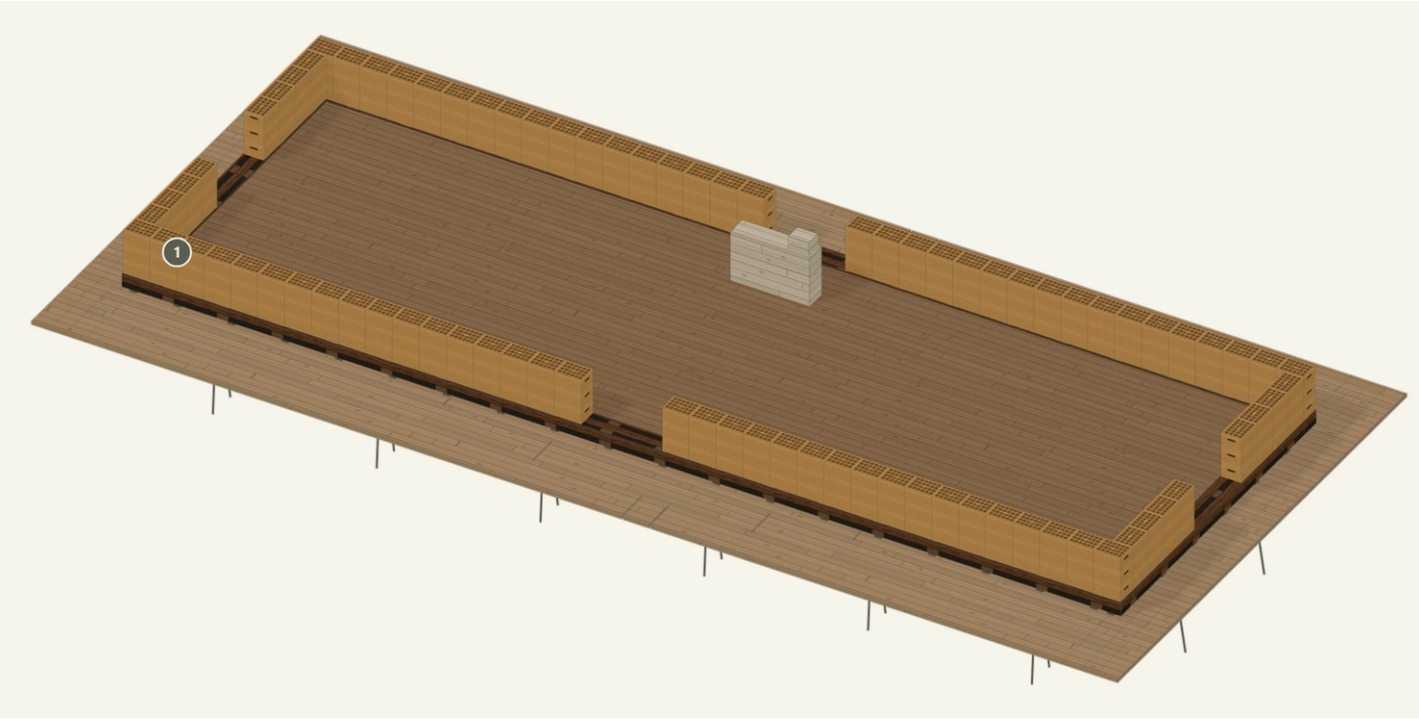
- 1 timber sleepers
- 2 steel U-channel
- 3 hardwood packing blocks



3 Step 2 Floor joists

Timber joists every 50 cm span across the steel channel; the plank floor is laid on them.

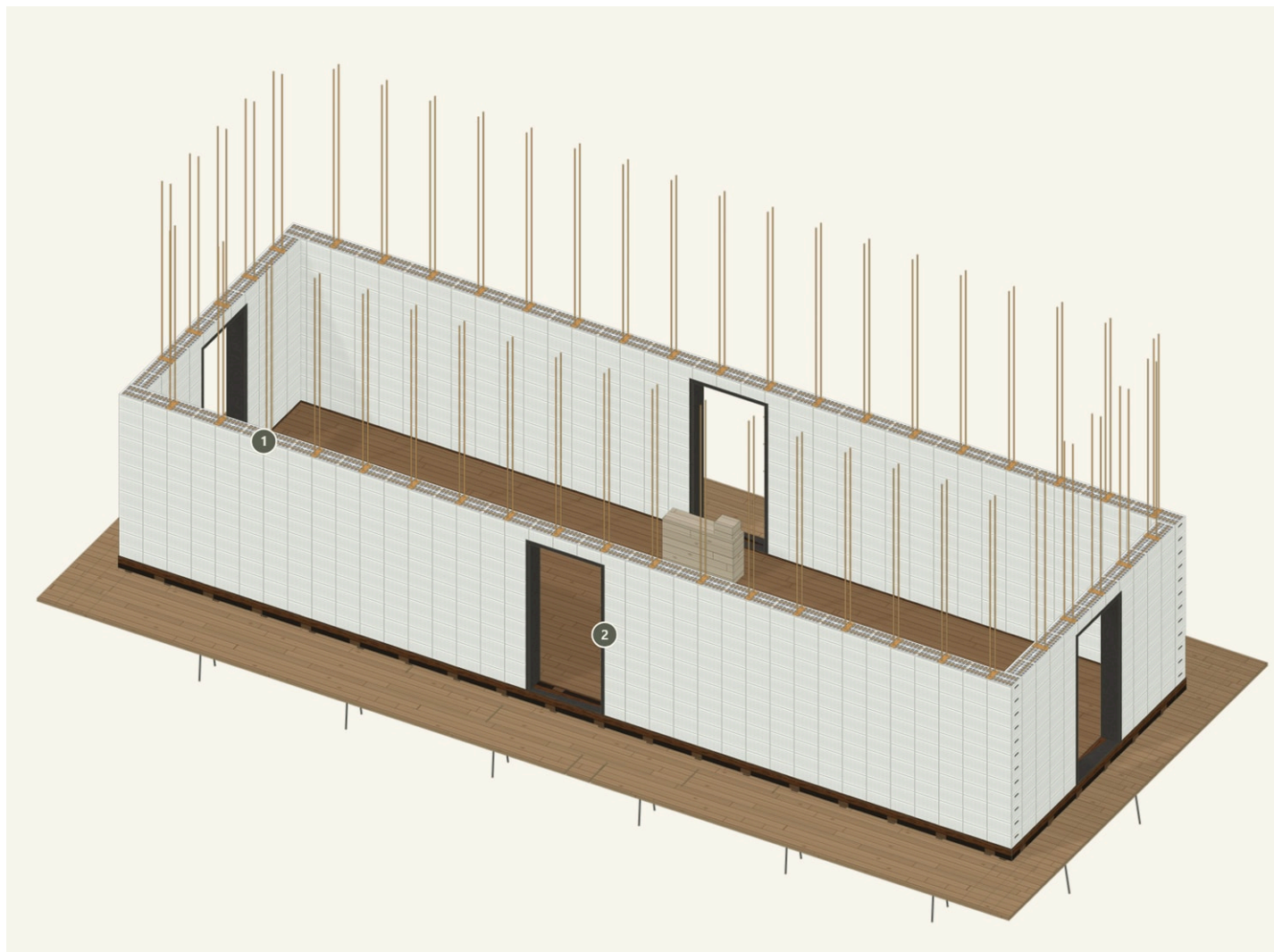
- 1 timber joists every 50 cm



4 Step 3 Floor and first courses

On the plank floor, standard Dutch beer crates (395 x 295 x 235 mm), a colourless translucent edition, are stacked in stack bond on the channel: long walls 37 crates, end walls 11.

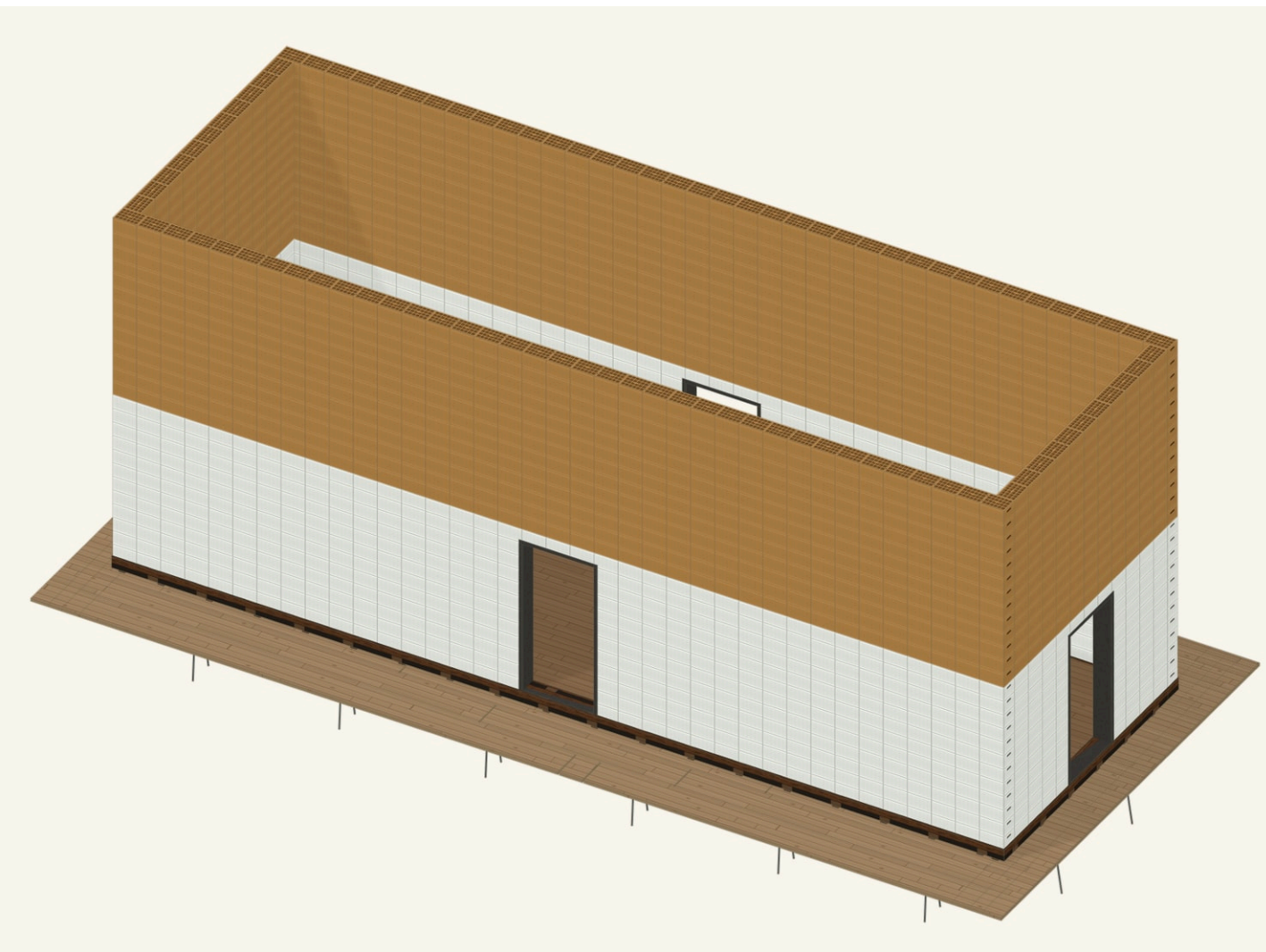
- 1 beer crate 395 x 295 x 235



5 Step 4 Tie rods

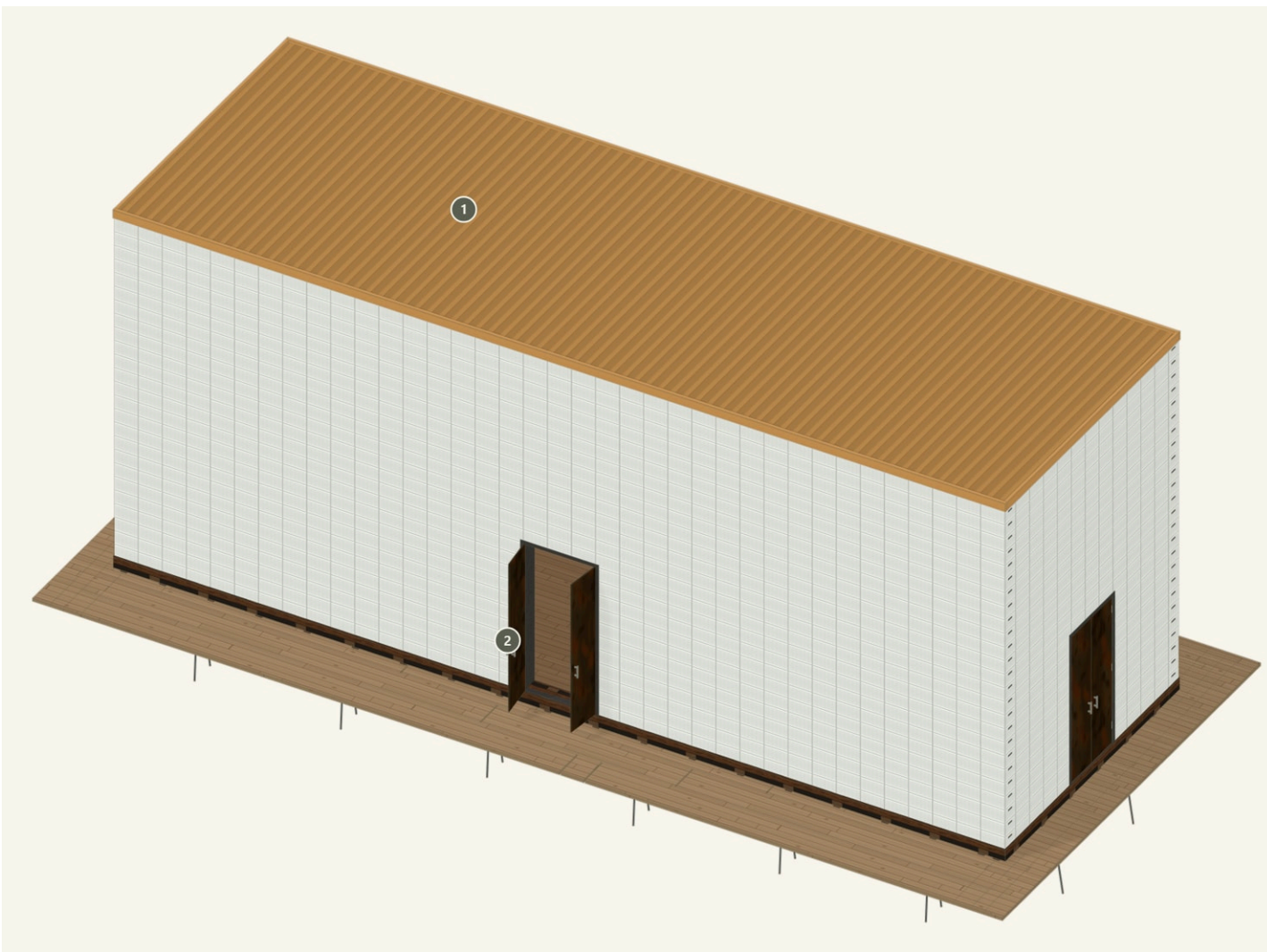
Every second course a pair of threaded rods with clamp plates; vertical rods every second crate post-tension the crate wall. Steel frames line the door openings through the full wall depth.

- 1 threaded rods + clamp plates
- 2 steel door frame, full wall depth



6 Step 5 Walls

25 courses, about 2,270 crates, give a 6 m high load-bearing translucent wall.



7 Step 6 Doors and roof

Rusty steel double doors on all four sides; a galvanised trapezoidal deck in a steel edge closes the box. Built by six people in one day.

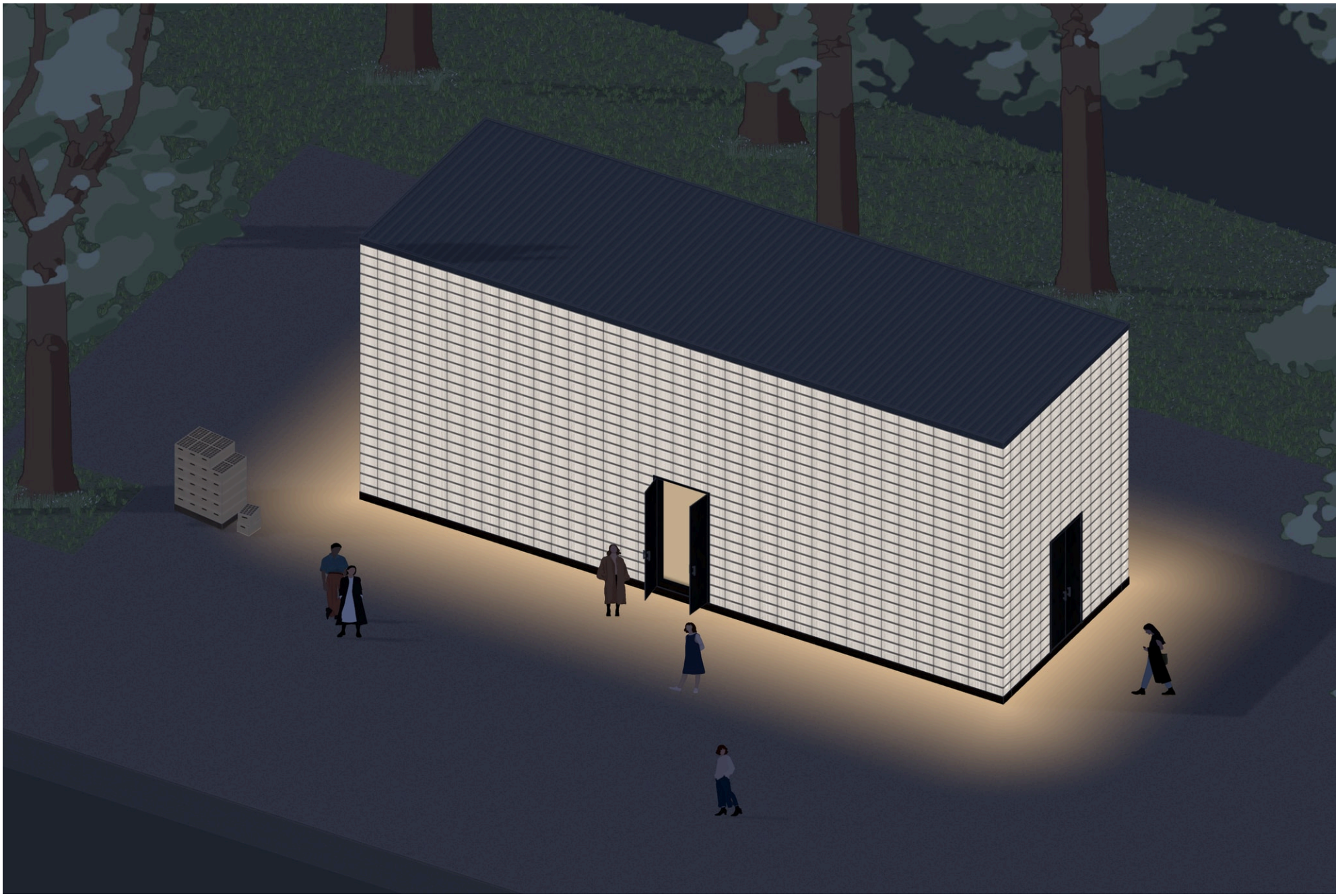
- 1 trapezoidal steel deck
- 2 steel double doors on all four sides



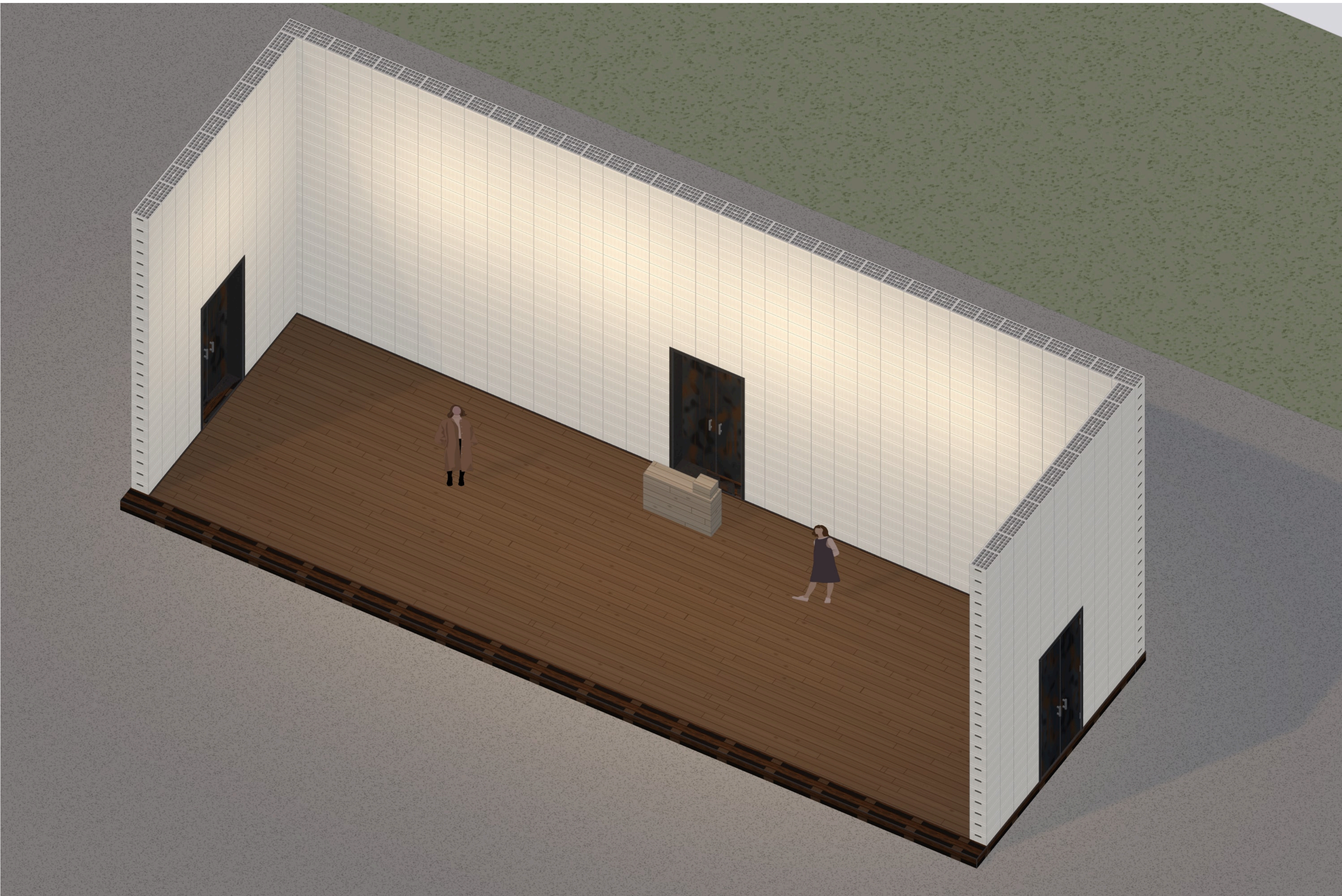
1 Day white crates in sun and tree shade



2 Dusk the crates glow from inside



3 Night light falls out onto the gravel

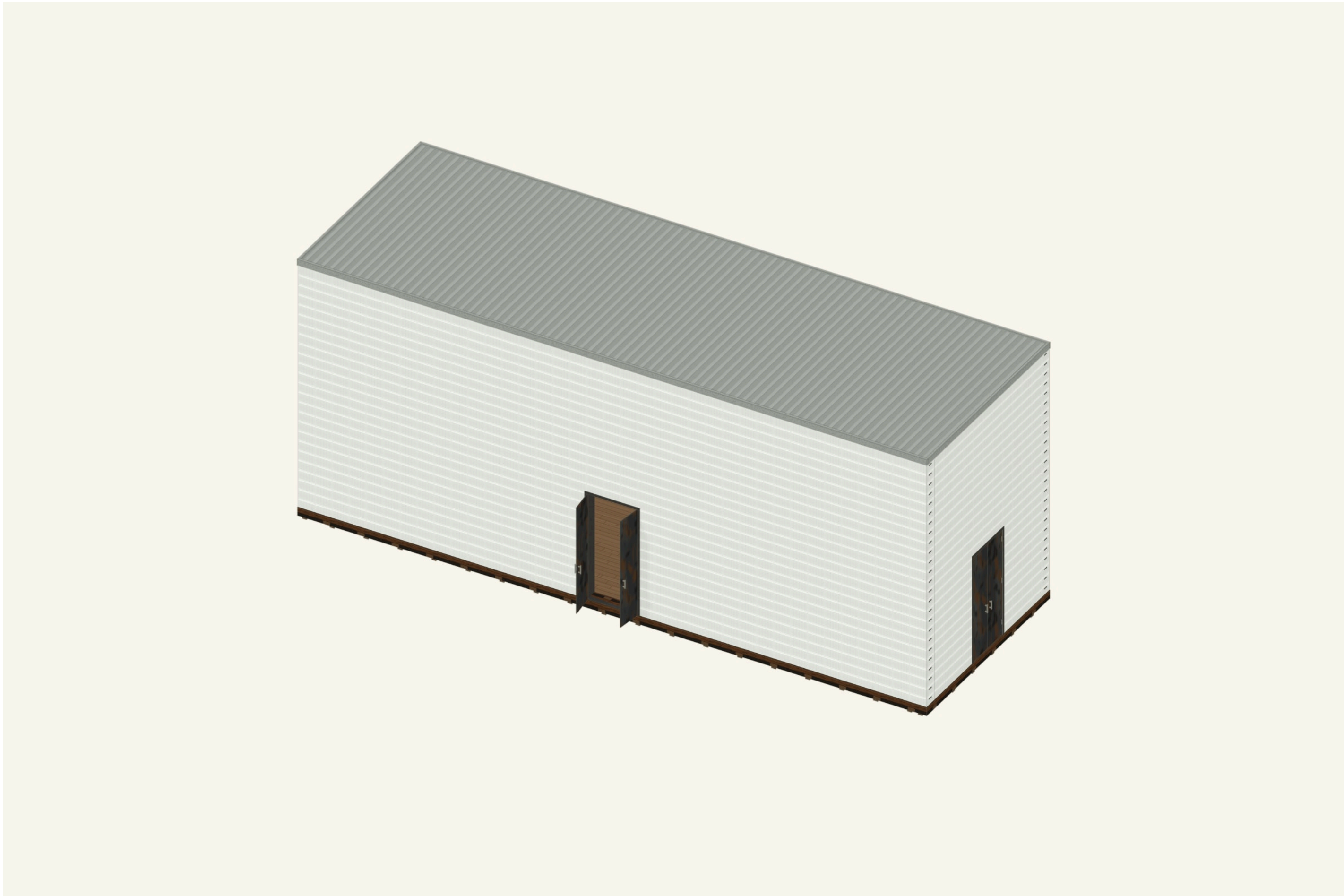


4 Inside roof and south wall lifted off - light through the crate walls

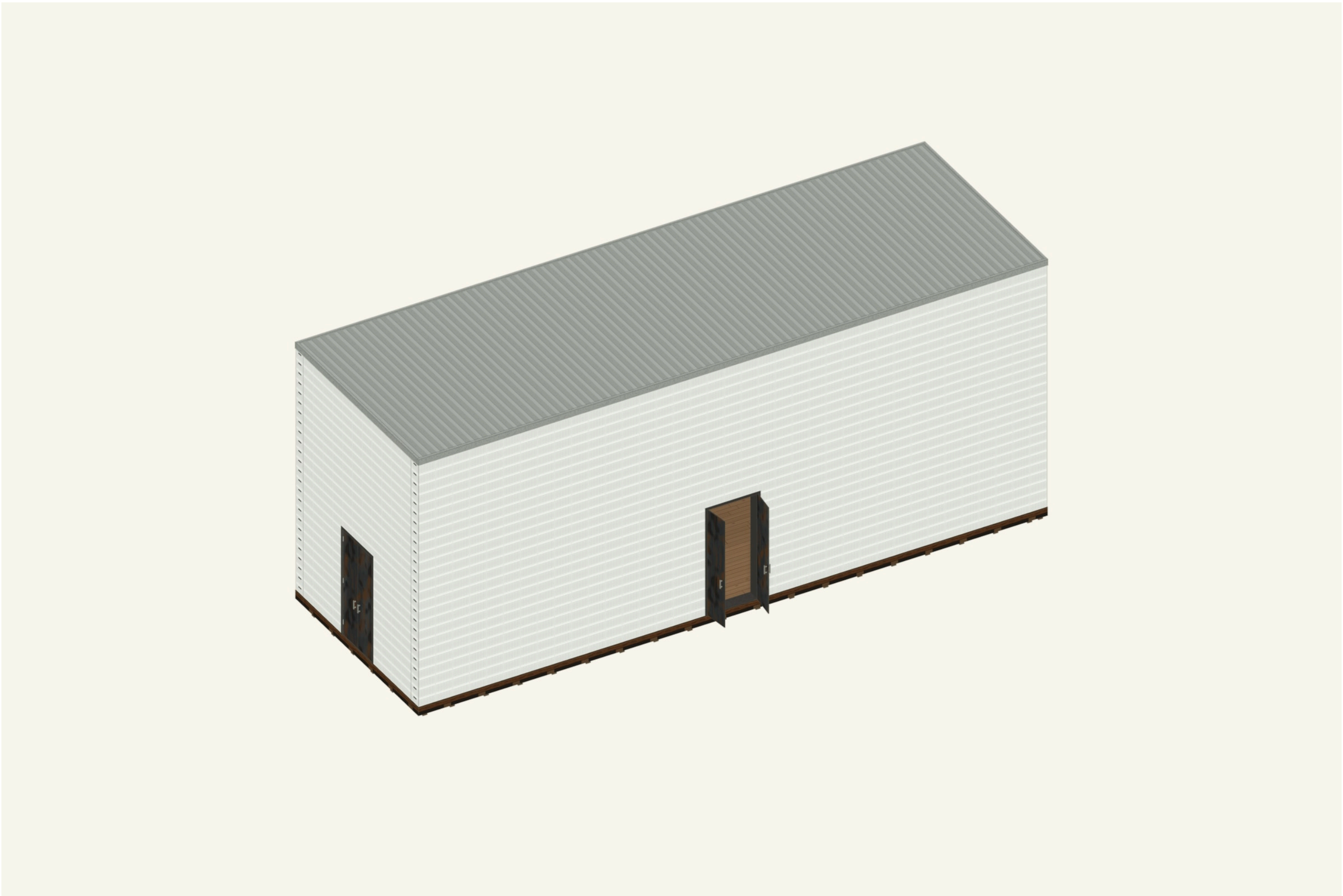
Light

The crates are translucent and load-bearing at once: by day the walls read white and fine-grained, at dusk and night the building turns into a lamp.

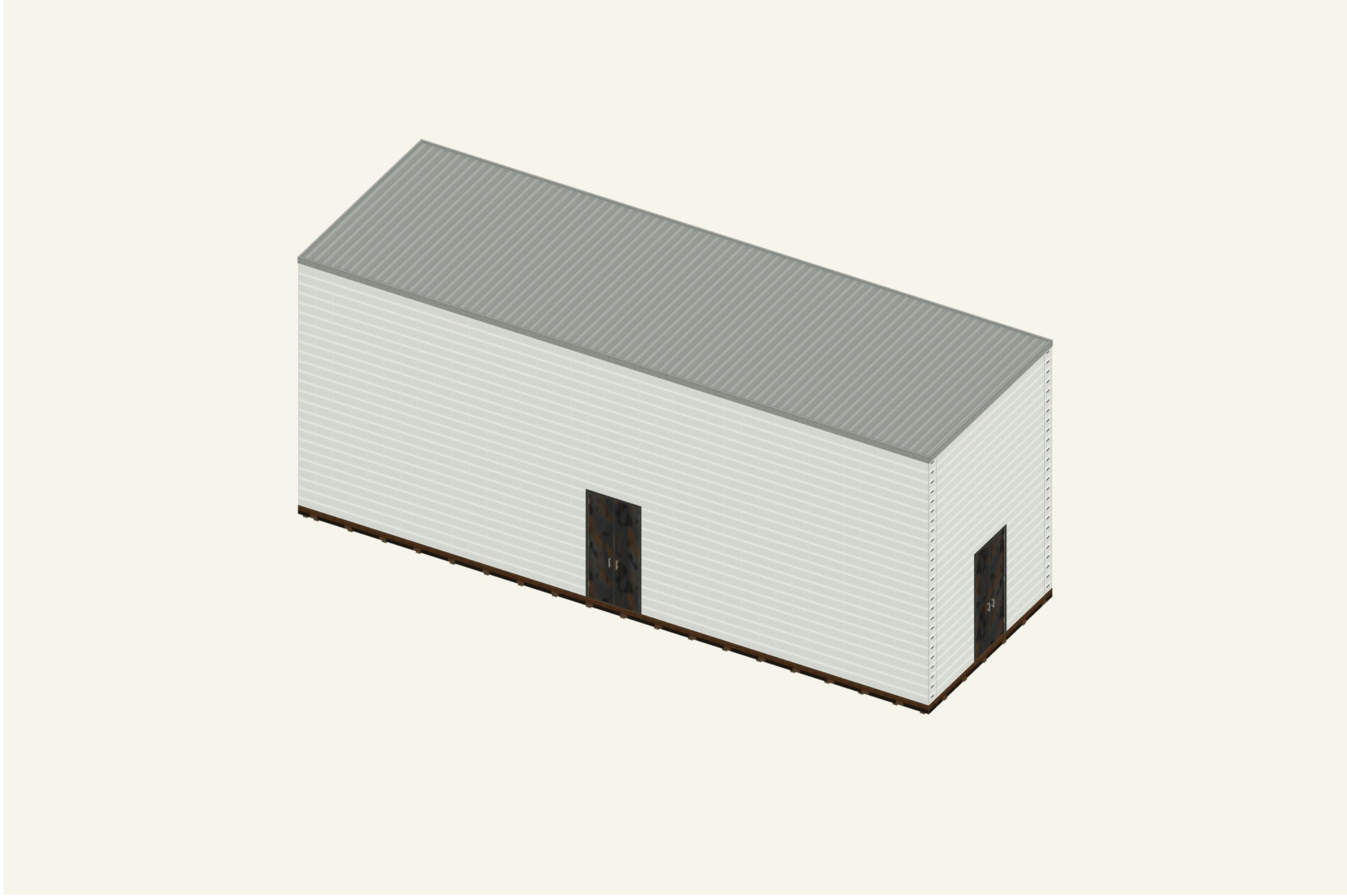
Day: sun from the south-west, tree shade on the long facade. Dusk and night: a soft white glow from all sides, no point lamps (they would shine through the walls), a light pool on the gravel.
The interactive viewer adds a dynamic glow with the slow shadows of people walking inside.



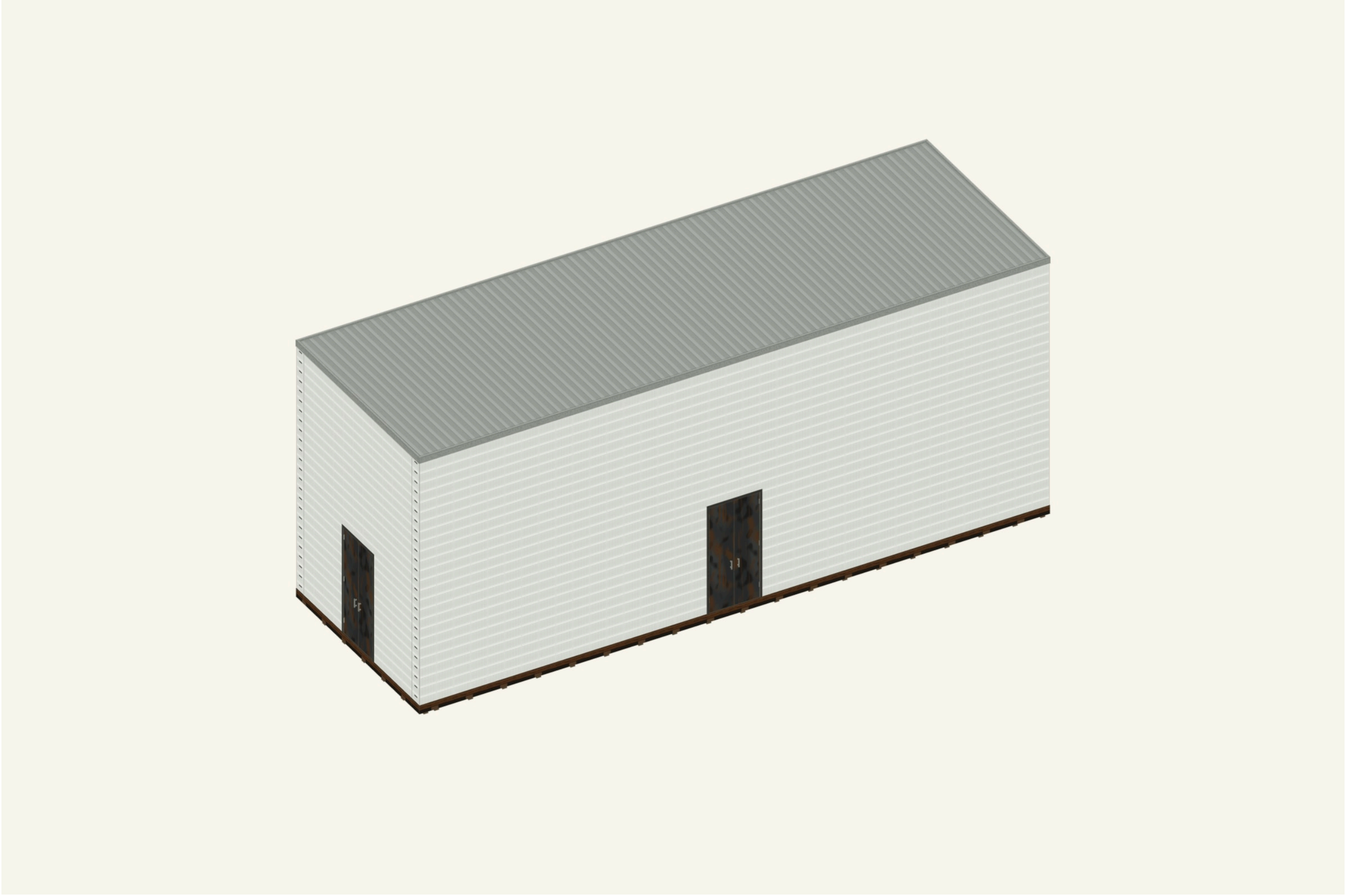
1 Only the building, from the south-east same cameras as sheet 01, without site, trees and people



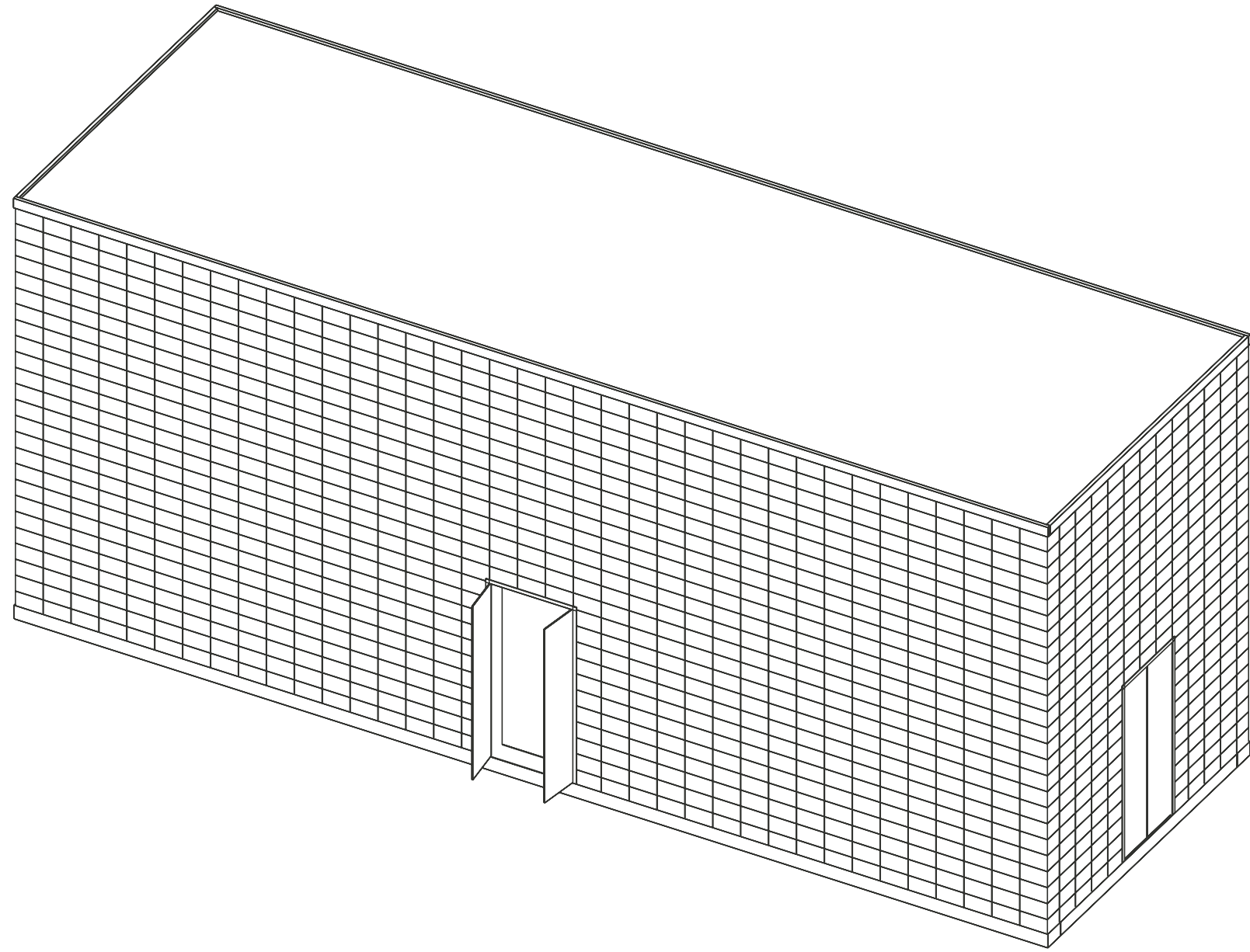
2 Only the building, from the south-west same cameras as sheet 01, without site, trees and people



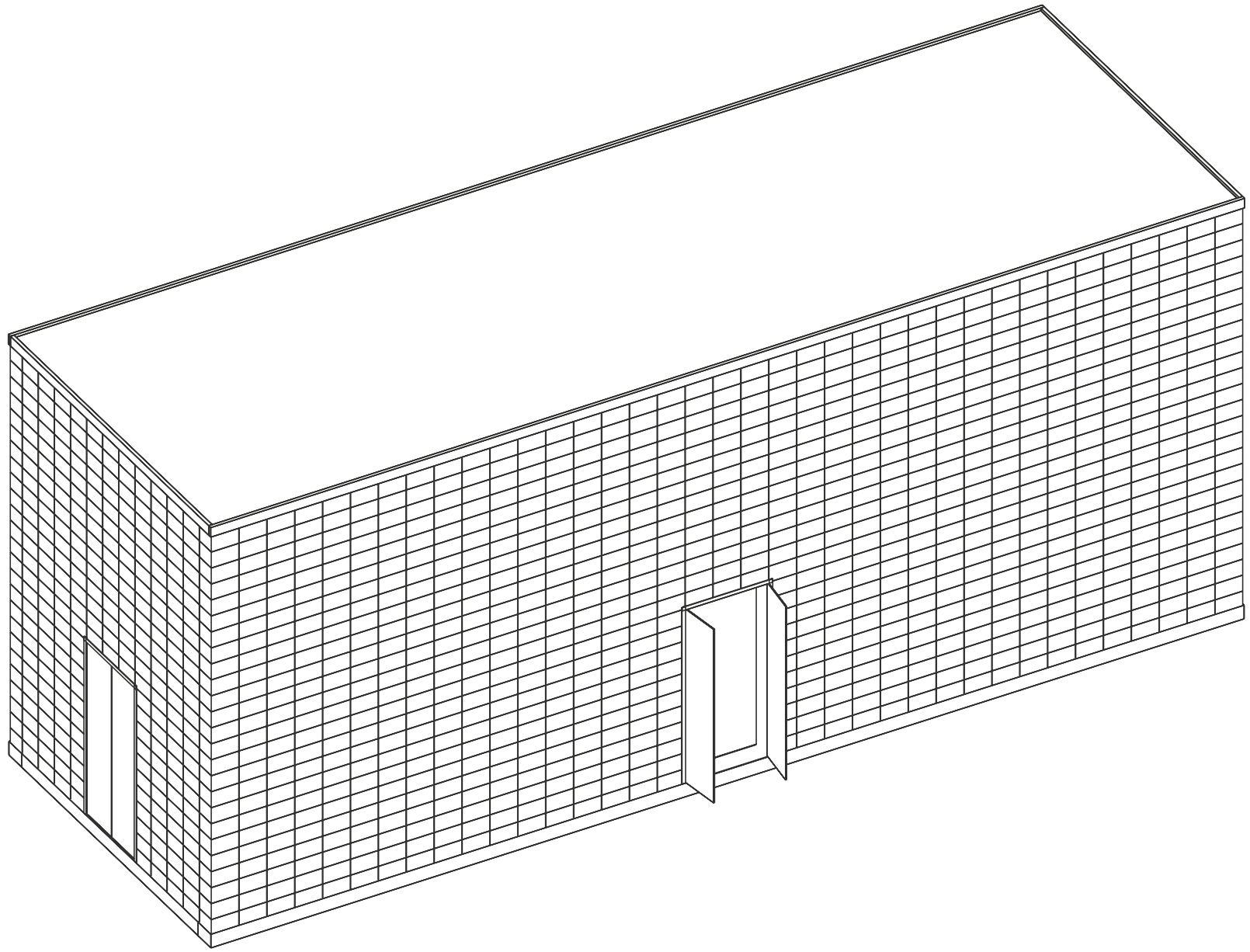
3 Only the building, from the north-west same cameras as sheet 01, without site, trees and people



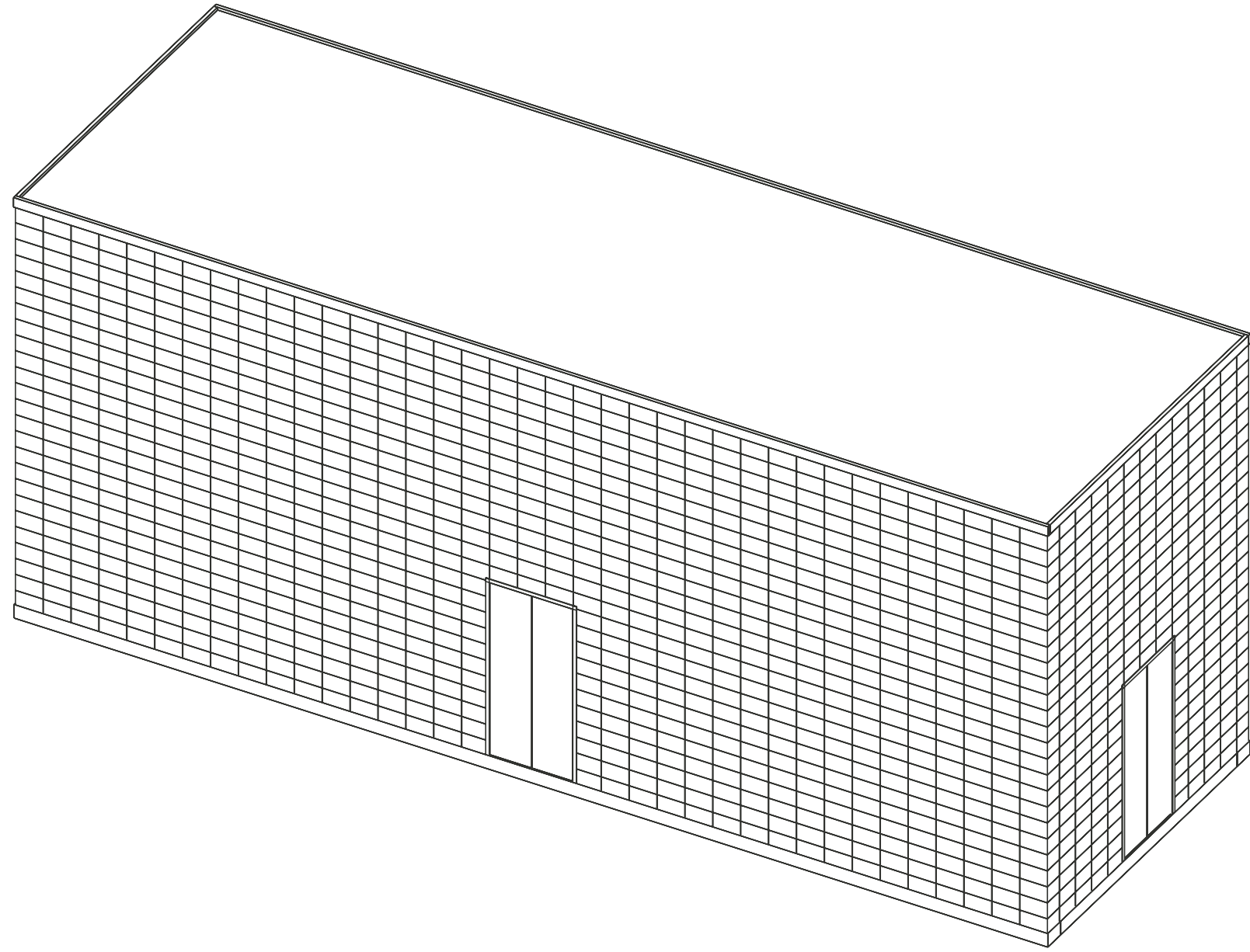
4 Only the building, from the north-east same cameras as sheet 01, without site, trees and people



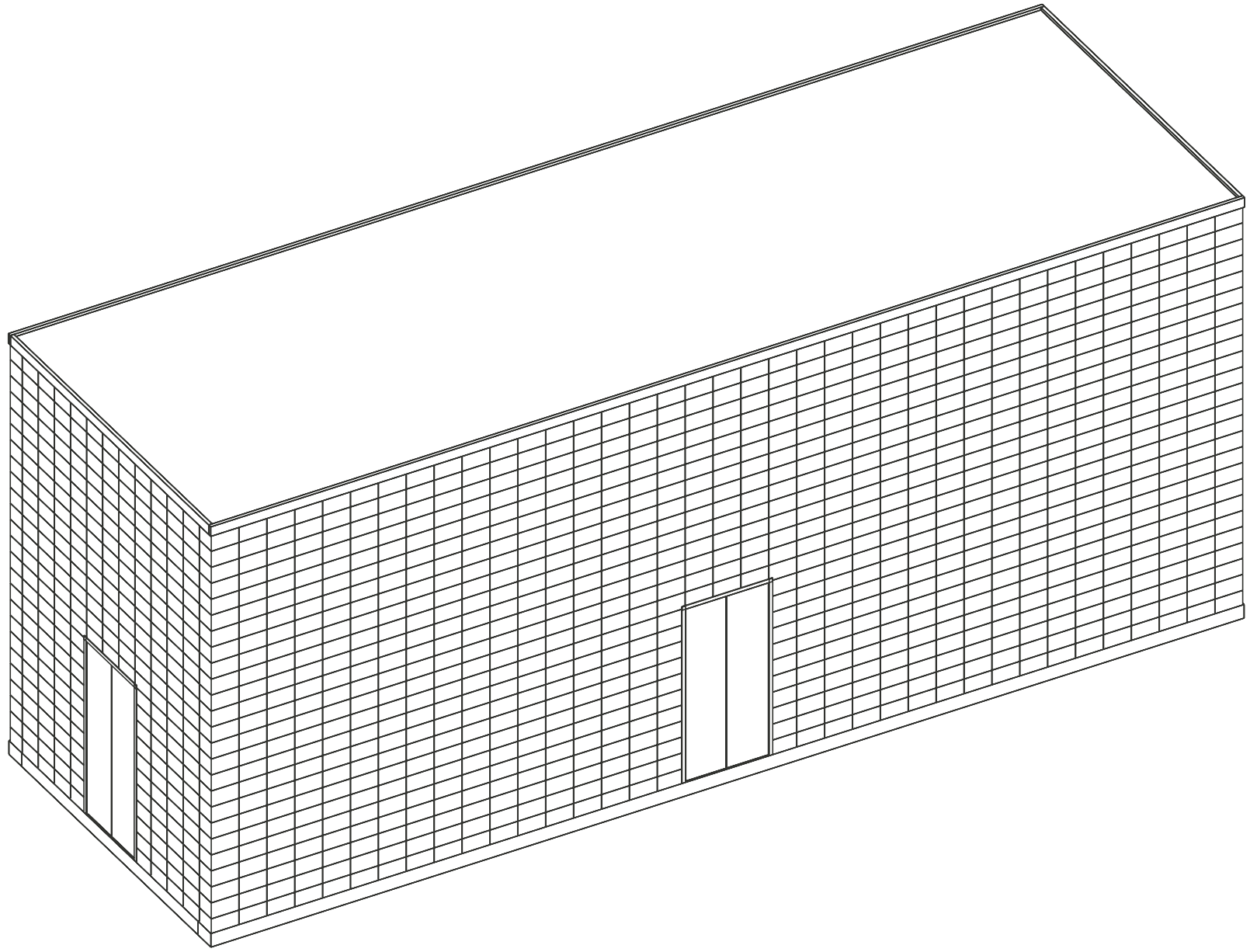
1 Ink axonometric from the south-east vector line drawing · the edges of all 2,268 crates



2 Ink axonometric from the south-west vector line drawing · the edges of all 2,268 crates



3 Ink axonometric from the north-west vector line drawing · the edges of all 2,268 crates



4 Ink axonometric from the north-east vector line drawing · the edges of all 2,268 crates